

Repository-scale metadata resource for reusable public proteomics

Ian Sitarik^{1,2,#}, Harikrishnan Ramadasan^{3,#}, Fatemeh Mirzadeh Sarcheshmeh^{4,#}, Eric W. Deutsch⁵, [contributors], Shomir Wilson⁶, Iddo Friedberg⁷, Wout Bittremieux^{4,%,*}, Tine Claeys^{3,%}

¹ Institute for Computational and Data Sciences, Pennsylvania State University, University Park, Pennsylvania, United States

² National Synthesis Center for the Emergence of Molecular and Cellular Sciences, Pennsylvania State University, University Park, Pennsylvania, United States

³ Department of Biomolecular Medicine, Ghent University, Ghent, Belgium

⁴ Department of Computer Science, University of Antwerp, Antwerp, Belgium

⁵ Institute for Systems Biology, Seattle, Washington, United States

⁶ Department of Human-Centered Computing and Social Informatics, Pennsylvania State University, University Park, Pennsylvania, United States

⁷ Department of Veterinary Microbiology and Preventive Medicine, Iowa State University, Ames, Iowa, United States

equal contribution

% equal contribution

* To whom correspondence should be addressed: wout.bittremieux@uantwerpen.be

Abstract

Public proteomics repositories contain a rapidly growing record of biological experiments, but reuse remains limited by incomplete, heterogeneous, and non-machine-readable metadata. Essential information about sample origin, disease context, experimental design, acquisition strategy, labeling, enrichment, and instrument configuration is often absent from structured repository fields or distributed across manuscripts, raw files, and free-text annotations. Here, we present HAMLET, a hybrid agentic metadata enrichment framework that reconstructs biological, experimental, and technical context for public mass spectrometry-based proteomics datasets. HAMLET integrates evidence-constrained text mining, raw data-derived technical inference, ontology normalization, and validation. Specialized language-model agents extract biological descriptors, technical parameters, and experimental design variables from open-access manuscripts, while raw file analysis infers acquisition mode, labeling evidence, organismal origin, and modification-associated signals directly from spectral data. Extracted values are integrated with supporting evidence, normalized to controlled vocabularies where possible, and retained with source attribution. Applied to 15,000 PRIDE datasets and integrated within ProteomeXchange, HAMLET generates a metadata-enriched resource that supports structured discovery for biological reanalysis, technical benchmarking, artificial intelligence training-data selection, and repository curation. By making dataset context computable, HAMLET provides infrastructure for systematic reuse of public proteomics data.

Introduction

The systematic sharing of biological data has become a central mechanism by which life sciences accumulate, validate, and extend knowledge. Large-scale community resources have enabled discoveries that could not have emerged from individual studies alone, covering genome annotation¹, single-cell atlases², structural biology³, and machine learning models trained on public molecular data^{4–6}. This shift makes data stewardship integral to scientific practice. The FAIR principles, which emphasize that data should be findable, accessible, interoperable, and reusable, provide a widely adopted framework for realizing this goal⁷. Across omics fields, public repositories now preserve increasingly large volumes of sequencing, transcriptomics^{8,9}, proteomics¹⁰, metabolomics^{11–13}, imaging¹⁴, and multi-omics data¹⁵. The scientific value of these resources, however, depends not only on whether files are deposited, but on whether they are accompanied by enough context to interpret and reuse them reliably. That context is itself a scientific output, one that can be incomplete at deposition and improved over time.

Metadata provides this contextual layer, describing the biological material, experimental design, analytical workflow, and technical conditions under which data were generated. It connects molecular measurements to organisms, tissues, cell types, disease states, perturbations, time points, replicate structures, instruments, acquisition methods, and processing choices. Without this information in a structured and computable form, deposited datasets remain technically accessible but scientifically opaque. This problem becomes more consequential as public datasets are increasingly reused for purposes not anticipated by the original investigators, including cross-study meta-analysis, benchmarking of computational methods, and training of artificial intelligence models. In each of these settings, incomplete or ambiguous metadata translates directly into inappropriate dataset selection, confounded comparisons, and limited reproducibility. Retroactive manual annotation of this metadata remains time-consuming and error-prone, and it often becomes interpretive as relevant facts are implied rather than stated.

Mass spectrometry (MS)-based proteomics illustrates both the promise and the difficulty of FAIR data reuse. Public proteomics repositories, coordinated through ProteomeXchange¹⁰, now contain large numbers of datasets that include raw instrument files, processed identifications, and links to associated publications. These datasets represent a substantial resource for biological discovery and method development supporting repository-scale reanalysis¹⁶, comparison of protein abundance across biological conditions^{17,18}, evaluation of search engines and quantification workflows¹⁹, and development of spectral prediction and *de novo* peptide sequencing models^{20,21}. Yet meaningful reuse, often with a biological goal, requires knowing the organism, tissue, cell type, disease state, sample preparation, digestion enzyme, labeling strategy, enrichment protocol, acquisition mode, instrument platform, fractionation scheme, post-translational modification focus, and replicate structure. Much of this information is absent from structured repository fields, inconsistently represented across submissions, or available only in free text. Most of these descriptions sit in the associated manuscript, so reuse is bounded in practice by whether manuscripts can be read at scale.

Community standards provide a necessary foundation for addressing this limitation. SDRF-Proteomics represents the relationships among samples, assays, raw files, and experimental variables in a structured tabular format²². Together with annotation tools that support standard-compliant submissions²³, it has improved the ability to describe new proteomics datasets in a

form that downstream computational workflows can consume¹⁸. However, this standard does not address the preexisting metadata gap. Indeed, many historical and valuable datasets were deposited before these standards were widely adopted, and even recent datasets often contain incomplete, inconsistent, or overly coarse annotations. In addition, important biological and experimental details are often described only in manuscripts, supplementary files, or repository free-text fields. Researchers who wish to reuse public proteomics data therefore still commonly inspect papers and project records by hand before deciding whether a dataset suits a specific analysis, a process that is both time-consuming and error prone.

No single information source is sufficient for this task, and integrating the available sources is the central difficulty. Reconstructing metadata requires drawing on four complementary streams. First, publications contain biological context, study rationale, sample descriptions, and experimental comparisons that are not encoded in raw instrument files. Second, raw MS files provide direct technical evidence about acquisition characteristics, scan structure, labeling signatures, peptide evidence, organismal origin, and modification patterns. Third, repository records provide accession-level context, submitted annotations, controlled vocabulary terms, and links between datasets and publications. Fourth, controlled vocabularies and ontologies provide the semantic layer required to make values comparable across studies. Integrating these streams into a single representation is what allows metadata reconstruction to be scalable, auditable, and robust to the heterogeneity of public proteomics data.

Here, we present HAMLET (Hybrid Agentic Metadata Literature Extraction and Technical annotation), an agentic framework that reconstructs the biological, technical, and experimental context of public MS-based proteomics datasets. HAMLET combines publication mining, raw data-derived technical inference, repository metadata integration, ontology normalization, and evidence-based validation to generate structured metadata records for PRIDE²⁴ and ProteomeXchange projects. Its text mining component uses specialized agents for biological metadata, technical metadata, and experimental design, with each extracted value linked to supporting evidence. Its technical component analyzes raw MS data to infer acquisition properties, labeling strategies, organismal origin, and modification-related metadata. Extracted values are normalized to controlled vocabularies and ontologies where possible, and conflicts among manuscript-derived, raw data-derived, and repository-derived evidence are retained with source attribution rather than being collapsed into a single unverifiable answer.

We evaluate HAMLET using expert-curated metadata annotations, existing SDRF-Proteomics records, and inter-annotator agreement among human curators which sets a realistic ceiling on what automated extraction can be expected to achieve. Applied to public proteomics projects with associated open-access publications, HAMLET generates a metadata-enriched resource that improves structured discovery across biological, experimental, and technical dimensions. We demonstrate how this resource supports novel approaches for data reuse, including searches for datasets suitable for artificial intelligence model training, biological reanalysis, technical benchmarking, and repository curation. By making the context of public proteomics datasets computationally accessible, HAMLET provides infrastructure for more systematic reuse of proteomics data and a general model for metadata enrichment in data-intensive molecular biology.

Results

Public proteomics metadata is incomplete, heterogeneous, and difficult to curate consistently

Structured, machine-readable metadata remain available for only a minority of public proteomics datasets (**Figure 1a**). Of the 39,494 PRIDE PXD accessions considered at the time of writing (July 2026), only 16,621 (42%) were linked to a publication indexed in PubMed Central (PMC) and 14,774 of these carry a reuse-permissive CC-BY or CC0 license that permits text mining. Structured SDRF-Proteomics metadata availability is even more rare: only 392 datasets, less than 1%, carry an SDRF record. More importantly, only 130 datasets are available in PMC, open licensed, and SDRF-annotated. Consequently, the set of datasets that are reusable at scale is 0.3% of all PRIDE datasets.

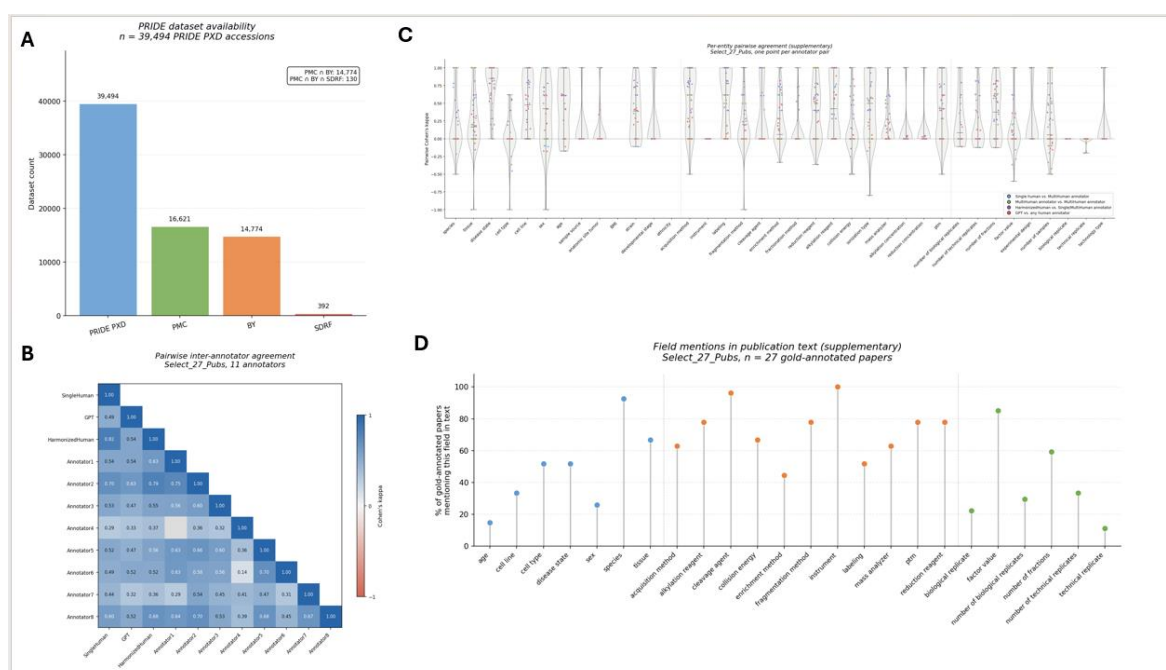


Figure 1. Public proteomics metadata are incomplete and heterogeneous, and annotator disagreement concentrates in fields that are rarely stated explicitly in the manuscript. (a) Availability of PRIDE datasets with corresponding linked publications (PMC), open-access full text (BY), and SDRF-Proteomics metadata (SDRF), across all 39,494 PRIDE PXD accessions considered. (b) Pairwise Cohen's κ across the 27-publication gold-standard set, pooled across all annotated fields: SingleHuman, HarmonizedHuman, and eight independent human annotators (Annotator1–8) against each other. (c) Per-entity pairwise Cohen's κ across the same annotator set, one point per annotator-pair comparison, grouped by biological, technical, and experimental design fields. (d) Percentage of the 27 gold-annotated papers mentioning each field in text, grouped by biological, technical, and experimental design fields.

Although the majority of metadata should be available within the associated publications, expert annotation showed that metadata recovery is not straightforward. We assembled a benchmark corpus during a hackathon at the EuBIC-MS 2025 Developers Meeting [REF], supplemented by a further curation and annotation effort performed carried out by a working

group supported by the National Science Foundation's National Synthesis Center for Emergence in Molecular and Cellular Sciences (NCEMS). This corpus covered 27 papers and 67 entities grouped into biological, technical, and experimental design metadata following SDRF-Proteomics consensus on entities.

Annotation was performed under three setups designed to separate three distinct sources of variability in the ground truth itself. First, a gold-standard subset of 27 publications was annotated independently by a team of eight EuBIC-MS experts (Annotator1 – Annotator8). Second, these annotations were reduced to a multi-human consensus by majority agreement per field (HarmonizedHuman), capturing the variability between experts reading the same manuscript. Third, the same 27 publications were re-annotated by a single NCEMS expert (SingleHuman), whose agreement with the consensus captures the variability within one consistent annotator and removes disagreement between humans. Pairwise Cohen's κ , pooled across all annotated fields (**Figure 1b**), showed that the single expert tracked the multi-annotator consensus closely ($\kappa = 0.82$ for SingleHuman vs. HarmonizedHuman), while agreement among the eight independent annotators themselves varied far more widely, from $\kappa = 0.14$ to $\kappa = 0.75$ depending on the pair. Furthermore, agreement is unevenly distributed across individual fields (**Figure 1c**). Breaking down pairwise Cohen's κ per entity shows a visible gradient: technical fields such as instrument, cleavage agent, and mass analyzer exhibit higher agreement than biological descriptors, which span a wider range of agreement depending on the pair. Experimental design fields show the lowest and most compressed agreement of the three classes, with several sitting at or below zero for multiple pairs.

Much of this field-level variation tracks how often a field is stated within the manuscript. Across the 27 gold-annotated papers, text-mention rates vary widely by field (**Figure 1d**): instrument, cleavage agent, and species are mentioned in nearly every paper, while fields such as age and technical replicate identifiers are mentioned in only a minority. The fields with the lowest text-mention rates are disproportionately the same experimental design fields that showed the lowest inter-annotator agreement. In other words, disagreement concentrates where the manuscript states the least, which is consistent with these fields requiring inference from narrative context rather than extraction of an explicitly reported value.

Together, these results show that the proteomics metadata gap is both structural and interpretive. Many public datasets lack structured metadata, and even when relevant information is present in manuscripts or repository records, recovering it requires source-aware interpretation, normalization, and validation. The interpretive component of this gap is not spread evenly across metadata categories, it concentrates in fields that are rarely stated explicitly in the text and on which even expert annotators disagree most. Metadata incompleteness is therefore not only a repository-entry problem; it is a scientific interpretation problem that requires evidence-aware extraction and validation. [REF

HAMLET integrates publication-derived, raw data-derived, and repository-derived evidence into structured metadata

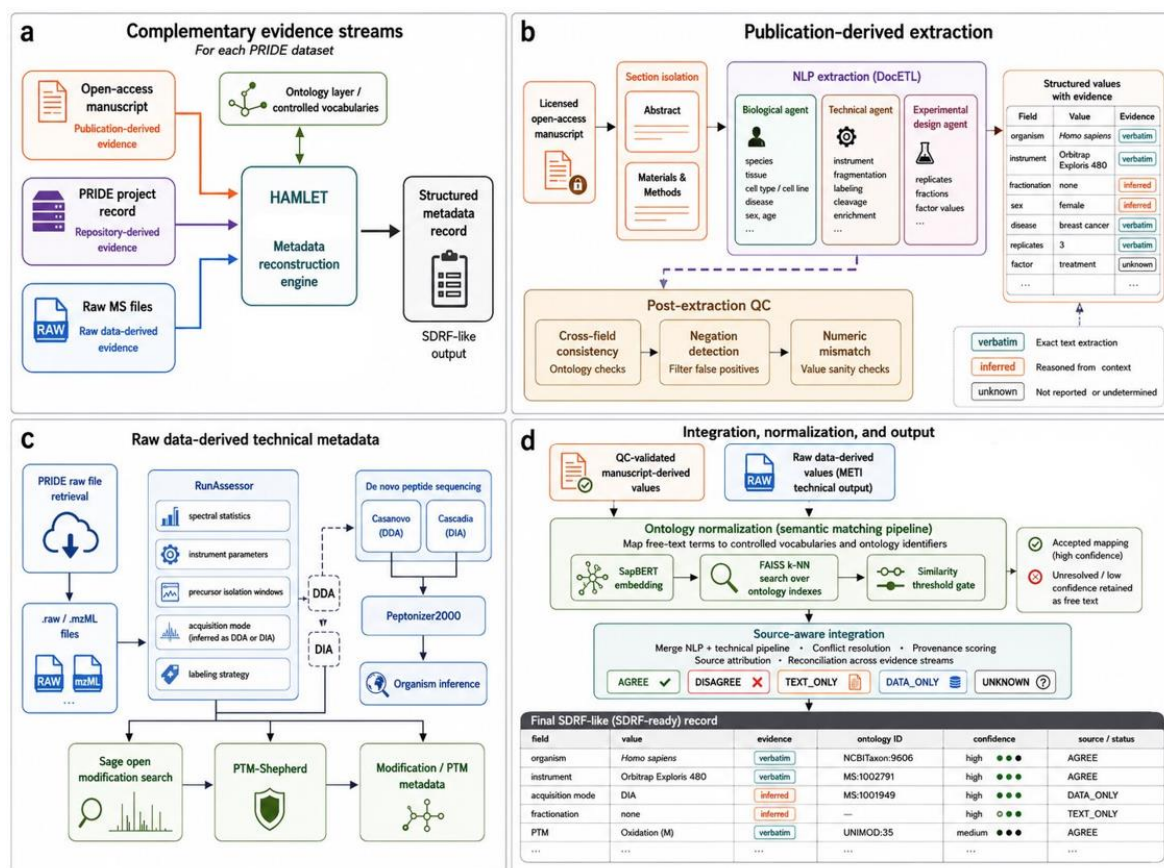


Figure 2. HAMLET integrates raw data-derived, publication-derived, repository-derived, and ontology-derived evidence into structured proteomics metadata. (a) HAMLET processes complementary inputs for each PRIDE dataset—the associated open-access manuscript (publication-derived evidence), the PRIDE project record (repository-derived evidence), and raw MS files (raw data-derived evidence)—harmonized through a shared ontology layer into a structured, SDRF-like metadata record. **(b)** Publication-derived extraction. The manuscript is reduced to its abstract and Materials and Methods sections and processed by three specialized natural language processing agents (biological, technical, experimental design) within the DocETL framework, producing structured field–value–evidence triples labeled verbatim, inferred, or unknown, and passed through post-extraction quality control (cross-field consistency, negation detection, numeric mismatch checks). **(c)** Raw data-derived technical metadata. Raw files are retrieved from PRIDE and processed by RunAssessor to infer acquisition mode (DDA/DIA) and labeling strategy; *de novo* peptide sequencing (Casanovo for DDA, Cascadia for DIA) feeds Peptonizer2000 for organism inference, while Sage open modification searching and PTM-Shepherd summarization derive PTM-related metadata. **(d)** Integration, normalization, and output. Validated manuscript-derived values and raw data-derived values are merged through a semantic ontology-normalization pipeline and source-aware integration, yielding a final SDRF-like record with per-field evidence type, ontology identifier, confidence, and source-agreement status (AGREE, DISAGREE, TEXT_ONLY, DATA_ONLY, UNKNOWN).

The HAMLET pipeline addresses these challenges by reconstructing proteomics metadata from complementary evidence streams, rather than any single source of information (**Figure 2a**). For each PRIDE dataset, HAMLET integrates the associated open-access manuscript, raw MS files, and PRIDE project records, harmonizing them through a shared ontology layer. This design reflects the distributed nature of proteomics metadata: biological context and experimental design are often described in publications, technical acquisition evidence can be

inferred from raw files, repository records provide submitted annotations and accession-level context, and ontologies make extracted values comparable across studies. The text mining component is organized into four sequential layers—extraction, normalization, integration, and validation—implemented within the DocETL framework [REF]. Each layer constrains the one before it: extraction proposes values, integration reconciles them against the other evidence streams, normalization grounds them in ontologies, and validation checks that every retained value remains supported by its source. The provenance and evidence guarantees described below are products of this layered design rather than properties of the language model alone.

First, the technical pipeline extracts metadata directly from raw MS data (**Figure 2c**). The RunAssessor tool²⁵ is used to derive information related to the MS acquisition mode and labeling strategy. Data-dependent acquisition (DDA) and data-independent acquisition (DIA) runs are distinguished from precursor isolation window characteristics. Peptide evidence generated using *de novo* peptide sequencing with Casanovo²¹ for DDA data and Cascadia²⁶ for DIA data is used for organism inference through Peptonizer2000²⁷, while fast open modification searching using Sage²⁸ and modification inference using PTM-Shepherd²⁹ are used to derive modification-related metadata.

Second, the extraction layer mines metadata from open-access manuscripts. Three large language model (LLM)-based specialized agents separately target biological metadata, technical metadata, and experimental design information (**Figure 2b**). Each extracted value is returned with supporting evidence, while unavailable fields are explicitly marked as unknown. Crucially, the agents distinguish values stated verbatim from values reached by inference: a verbatim value must appear as an exact substring of its evidence sentence, whereas an inferred value carries an evidence string prefixed with “inferred:” followed by the sentence that licensed the inference. For example, a species recorded as “Homo sapiens” because patient samples are mentioned is retained as an inferred value rather than an explicit value. This separation makes the basis of every value auditable and lets downstream checks apply the appropriate grounding test to each kind of evidence. Post-extraction quality control checks structural rules followed by a conditional gleaning step when a valid value is missing in its evidence. This enforces that values are not retained without traceable provenance and makes evidence pairing an inherent part of the output.

Third, the integration layer combines repository-derived metadata, such as PRIDE project-level annotations and controlled vocabulary terms, with the publication- and data-derived evidence (**Figure 2d**). This integration follows a source-aware hierarchy in which raw data-derived evidence is prioritized for directly measurable technical fields, PRIDE properties are retained as repository-level evidence, and manuscript-derived values provide biological and experimental context. Agreement across sources is encoded using explicit labels: AGREE, DISAGREE, LLM_ONLY, RUNASSESSOR_ONLY, or UNKNOWN.

Fourth, the normalization layer grounds extracted values in controlled vocabularies and ontologies following SDRF-Proteomics guidelines³⁰ where possible, mapping species, tissue, cell type, cell line, disease, instrument, labeling strategy, fractionation, post-translational modifications, and sample source to stable ontology identifiers. Grounding is more than surface formatting: by resolving free-text terms to ontology nodes, HAMLET places each value in a hierarchy that makes it comparable across studies and exposes relationships between

co-extracted fields, so that a value is interpreted in light of prior structured knowledge rather than taken at face value. A term that cannot be resolved above a confidence threshold is retained as free text and marked as unnormalized, keeping the distinction between a grounded value and an ungrounded one explicit in the record.

The final output is a structured per-dataset record holding, for each field, the extracted value, its supporting evidence, an ontology identifier, confidence information, and source attribution. The record is SDRF-like rather than a complete SDRF-Proteomics file because text mining often cannot resolve information at the individual MS run level. Instead, when finer granularity is unavailable, HAMLET reports the relevant metadata at the experiment or project level.

Raw mass spectrometry files deterministically recover technical metadata

Technical metadata extraction was evaluated using a subset of 280 heterogeneous PXD datasets. First, RunAssessor summarizes MS-level distributions, precursor isolation characteristics, fragmentation-related properties, and labeling-associated signatures across the technical QC subset (**Figure 3a**). The acquisition mode resolves overwhelmingly to DDA, with only a small minority of runs classified as DIA, unknown, or ambiguous, consistent with DDA's continued dominance among public repository submissions. Fragmentation is dominated by high-resolution HCD, MS2 accounts for approximately 80% of scans, and precursor charge states concentrate at 2+ and 3+. Precursor and fragment mass tolerances cluster tightly, with fragment tolerance around 10 ppm for most runs, and labeling-related signatures are dominated by the unknown/undetermined category, reflecting how rarely raw spectral evidence alone yields an unambiguous labeling call. Together, these distributions show that RunAssessor recovers a coherent, internally consistent technical profile for the large majority of datasets rather than a scatter of unresolvable edge cases.

Second, raw files support organism inference independent of manuscript or repository annotations (**Figure 3b**). HAMLET generates peptide evidence using Casanovo and Cascadia for *de novo* peptide sequencing of DDA and DIA data, respectively, after which Peptonizer2000 was used for taxonomic assignment [REFs]. Across the 268 PXDs with sufficient peptide evidence for evaluation; recall, precision, F1, and balanced accuracy each reached or exceeded 0.77 for approximately 74% of datasets. The distribution of F1 scores was bimodal rather than centered on a single moderate value (mean F1 = 0.77, 95% CI $\pm$ 0.05): the large majority of PXDs achieved near-perfect organism recovery, while a smaller subset failed almost completely. This pattern indicates that organism inference from *de novo* peptide evidence succeeds or fails as a function of specific dataset properties, including closely related taxa with shared peptide sequences and datasets with limited high-confidence peptide evidence, rather than degrading gradually across the board.

Third, open modification searching recovers post translational modification (PTM)-related metadata directly from spectra (**Figure 3c**). Sage and PTM-Shepherd identified recurrent mass shifts corresponding to common modification classes across the 280 PXDs evaluated. Cysteine alkylation was detected in nearly every PXD, consistent with its origin as a near-universal sample-preparation artifact rather than a biological signal. Oxidation and deamidation followed a similar pattern, each appearing in roughly 40% of PXDs. Isobaric and

metabolic labeling showed the opposite relationship between prevalence and occupancy: TMT/isobaric labels and SILAC labels were each detected in only a small minority of PXDs, but where detected, their mean occupancy was the highest of any modification class, reflecting how labeling reagents, once used, are incorporated near-stoichiometrically rather than at a low background rate. Presence and occupancy therefore capture different aspects of a modification's role in a dataset: a modification detected in nearly every PXD can still be a minor background signal, while a modification detected rarely can dominate the labeled peptide population wherever it occurs. While these results indicate whether a modification signal was present in the data, they do not by themselves distinguish whether the modification reflects a biological PTM, an artifact introduced during sample preparation or analysis, or the consequence of a targeted enrichment strategy. HAMLET therefore records open-search modification evidence as observed PTM presence rather than as direct evidence of PTM-focused experimental design. Complementary evidence for specific enrichment or labeling modes is derived by RunAssessor where supported, including signatures consistent with phosphorylation enrichment and TMT labeling.

Taken together, these results show that raw files recover technical metadata that is often missing, ambiguous, or inconsistently reported in repository records and manuscripts, and that this recovery is internally consistent: acquisition and fragmentation parameters resolve cleanly for the large majority of datasets, organism inference succeeds or fails in a bimodal rather than gradual pattern tied to identifiable properties of the peptide evidence, and modification recovery separates how often a signal is present from how much of the proteome it affects. This data-derived extraction cannot supply the full biological or experimental context of a study, but it provides reproducible, source-independent evidence for the metadata fields that are written directly into acquisition records and spectra.

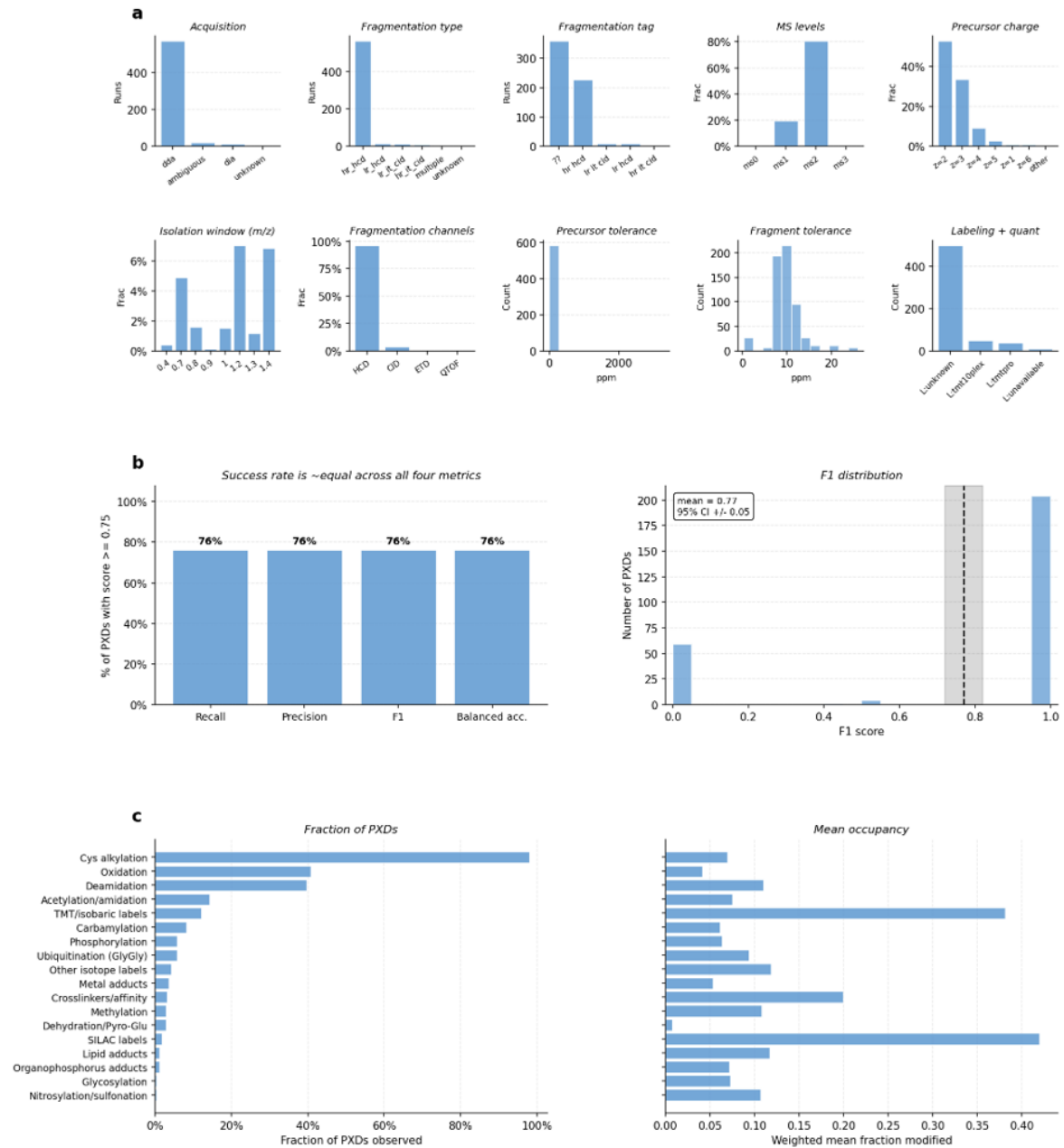


Figure 3. Performance of raw data-derived technical metadata. (a) RunAssessor-derived technical summaries across sampled datasets (n = 280 PXDs), including MS-level distributions, precursor isolation characteristics, DDA/DIA classification, fragmentation-related properties, and labeling/quantification signatures. (b) Organism inference from *de novo* peptide predictions (Casanovo for DDA, Cascadia for DIA), filtered for confidence and contaminant sequences and analyzed with Peptonizer2000, across n = 268 evaluable PXDs. Left: fraction of PXDs reaching a score ≥ 0.75 for recall, precision, F1, and balanced accuracy. Right: distribution of F1 scores (mean = 0.77, 95% CI ± 0.05). (c) PTM information recovered by open searching (Sage, PTM-Shepherd) across n = 280 PXDs. Left: fraction of PXDs in which each modification class was detected. Right: mean occupancy (weighted mean fraction of modified residues) where detected.

Publication-derived metadata extraction recovers biological context and experimental design

Because many metadata fields required for reuse are not encoded in raw MS files and must instead be recovered from manuscripts, HAMLET uses evidence-grounded LLM agents to extract biological context, technical information, and experimental design variables from associated open-access publications.

Prompt optimization and pipeline development of HAMLET were performed using 137 PXD datasets with SDRF-Proteomics annotations that were either available through the BigBio GitHub repository or generated manually through the EuBIC-MS annotation effort described previously. These datasets were divided into development and held-out test sets using an approximately 80/20 split, comprising 107 and 30 datasets, respectively. All results reported below were obtained exclusively on the 30 held-out test datasets and performance was calculated against the gold-standard SDRF-Proteomics annotations.

The decisive factor in this extraction was the framework around the model rather than the model itself. Comparing a single general master prompt against the specialized agents shows this most clearly in terms of consistency rather than accuracy alone (**Figure 4a**): across a 30-PXD test set with a known manually annotated gold standard, the master prompt extracted 11 ± 10 fields per PXD with Llama and 148 ± 60 with GPT, a thirteen-fold difference in yield between models and a within-model spread wider than the GPT mean itself. The specialized agents removed this variability entirely, returning exactly 41 fields per PXD regardless of model (41 ± 0 for both Llama and GPT), because the agent schema fixes the set of fields returned and requires every field to be populated, with a value or with unknown, rather than leaving output volume to the model's own judgment. Decomposing the task therefore raised average performance and more importantly, it converted an unpredictable, model-dependent output into a fixed, auditable one, a precondition for treating the result as structured metadata at all rather than as free-form text with variable coverage. Prompt design mattered as much as this decomposition: agent-specific field definitions, extraction rules, examples, and explicit instructions for unknown values improved precision and reduced over-inference. Against these gains, the choice of model was secondary. Benchmarked across six models on the same 30-PXD test set (**Figure 4d**), the open-weights Gemma 4 31B reached an overall F1 on par with the best-performing commercial model tested (GPT-5.4), ahead of Claude Opus 4.5, Gemini 3.5 Flash, and Qwen 3.7 Max, while Llama-4 Scout consistently trailed the other five models across every metadata class, most markedly on experimental design. That an open-weights model matched or exceeded several commercial systems once placed inside the same grounded, validated framework is the clearest evidence that the framework, not the model, carries the performance. Gemma 4 31B was therefore selected for large-scale processing, pairing competitive extraction performance with the cost advantage of local, open-weights deployment.

Extraction accuracy depended on the metadata category, though not entirely in the direction the original difficulty ranking assumed. Technical fields were the most model-invariant category in the multi-model benchmark (**Figure 4d**): every model converged on a similar F1 score regardless of its overall capability, consistent with these fields being stated in standard, formulaic Methods-section phrasing. This confirms the previously observed higher human

annotator agreement. Biological fields showed a moderate spread across models. Experimental design showed the widest spread of the three classes, with the strongest models reaching their highest F1 of any category there and the weakest model reaching its lowest: this category is the most sensitive to the quality of the extraction framework, rather than uniformly the hardest in absolute terms once a competent model and framework are used. Experimental design fields nonetheless remain structurally the hardest to extract in the sense that matters for design, since relevant information is usually implicit, distributed across manuscript sections, or expressed using study-specific descriptions rather than stated verbatim.

To ensure provenance of the metadata, each extracted value was linked to its supporting manuscript evidence. Directly stated values were returned with their source sentence, whereas inferred values were explicitly marked as inferred and paired with the sentence supporting the inference. Unsupported fields were returned as unknown rather than populated with plausible but unverified values. This evidence constraint increased precision and made missingness explicit, which is essential for a metadata resource intended to support dataset selection and reuse.

Grounding extracted values in ontologies is what turns this output from text into queryable data, and it also changes how performance must be measured. Equivalent metadata values are often expressed differently across manuscripts: “human” and “Homo sapiens” denote the same organism, and cell line aliases should resolve to a single stable identifier. Because exact string-matching underestimates correct recovery, we evaluated performance with a five-level hierarchical matching strategy that applies increasingly permissive criteria, from exact and normalized string matching through ontology-identifier matching to ancestor-based hierarchical matching and SciBERT-derived semantic similarity (**Figure 4b**).

Moving from exact to semantic, ontology-aware matching raised the F1 score for Gemma 4 31B on the 30-PXD benchmark from 0.424 to 0.940, with the largest absolute gain in the biological class (F1 rising from 0.350 to 0.942) and smaller, though still substantial, gains in technical (0.421 to 0.921) and experimental design (0.500 to 0.958) fields. Precision improved more than recall in every class (overall precision reached 0.997 against a recall of 0.891), indicating that most of the exact-match penalty came from values that were correct but expressed differently rather than from values that were absent or wrong. The largest gain concentrated in biological fields precisely because organism, tissue, and disease terminology carries the most synonymy and the most granularity mismatch between manuscript wording and ontology structure, so this is where semantic matching corrects the most measurement, not where the model performs the most poorly. The same grounding that corrects the measurement also builds the resource: by converting free-text values into stable ontology identifiers, HAMLET makes metadata directly comparable and queryable across studies. Each normalized value retained the original extracted text, supporting evidence, ontology identifier, canonical term, and confidence score. Values below the normalization threshold were retained as free text and marked as not normalized, so that uncertain mappings were never silently treated as curated annotations.

Ontology normalization corrects how a value is expressed, but it cannot supply a value the manuscript never states. Comparing coverage with and without the raw-data-derived pipeline (**Figure 4c**) shows where this ceiling falls: LLM-only extraction reached non-null coverage of

100% for species and technology type, fields that are almost always stated in the manuscript or already present in the PRIDE record, but only 93% for mass analyzer and instrument, and 73% for PTM evidence drawn from search results, fields that Methods sections frequently omit or describe too vaguely for text mining to resolve. Supplementing the LLM output with raw-data-derived information closed most of this gap, raising instrument coverage to 97%, mass analyzer to 100%, and PTM coverage to 83%. The residual PTM gap is expected rather than a shortcoming of either pipeline, since not every dataset carries a detectable open-search modification signal.

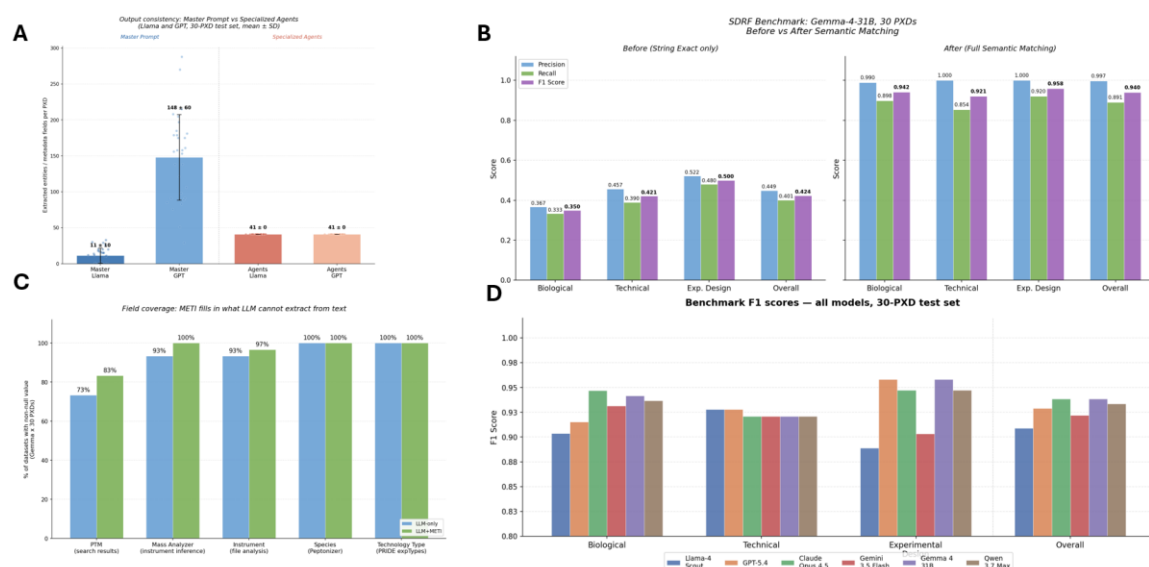


Figure 4. Publication-derived metadata extraction and validation. (a) Output consistency of a single master prompt versus the specialized agents, across two backend models (Llama, GPT), measured as extracted fields per PXD on a 30-PXD test set (mean $\pm$ SD). (b) Effect of the five-level hierarchical matching strategy on measured extraction performance for Gemma-4-31B on the same 30-PXD test set, before (exact string matching only) and after (full semantic matching), by metadata category. (c) Field-level coverage (% of datasets with a non-null value) achieved by LLM-only extraction versus LLM extraction supplemented with the raw-data-derived pipeline. (d) Benchmark F1 scores across six LLMs (Llama-4 Scout, GPT-5.4, Claude Opus 4.5, Gemini 3.5 Flash, Gemma 4 31B, Qwen 3.7 Max) on the 30-PXD test set, by metadata category.

Integration and LLM-as-a-judge

We evaluated how the integration of repository-derived metadata affected HAMLET performance by comparing the final annotations against the corresponding SDRF-Proteomics gold standard. Of the 30 evaluated PXDs, 26 contained METI metadata that could be integrated. For 20 of these datasets, integration altered at least one final HAMLET annotation and reduced agreement with the SDRF-Proteomics reference. During integration, METI metadata annotations were given priority over conflicting LLM-derived annotations because they were assumed to represent structured information directly associated with the deposited data. However, the observed performance decreases show that this assumption does not hold uniformly across all metadata fields. **Figure 5a** summarizes the types of annotation changes introduced during integration. In 127 cases, HAMLET assigned a value to a field that was not annotated in the corresponding SDRF-Proteomics file. These cases were excluded from the precision, recall, and F1 calculations because the available reference did not allow the

HAMLET annotation to be classified as either correct or incorrect. They may therefore represent either valid metadata enrichment or unsupported annotations. These cases were evaluated separately using the LLM-as-a-judge approach described below. Among the 77 evaluable performance decreases, species misidentification was the most frequent cause, accounting for 34 cases. Additional decreases occurred when multi-valued LLM annotations were reduced to a single METI value, thereby removing potentially relevant granularity. Instrument annotations were also affected: integration occasionally replaced specific instrument information with a more generic value or removed the annotation entirely. Together, these results indicate that METI integration can replace a correct and specific LLM-derived annotation with a less informative or conflicting value. The largest average reductions in field-level performance were observed for experimental-design annotations, particularly biological replicates, technical replicates, and fractionation information (**Figure 5b**). Substantial decreases were also observed for instrument and species annotations. These findings further highlight the difficulty of consistently extracting and reconciling experimental-design metadata across heterogeneous sources. To investigate these changes in more detail, we compared the five-tier match classifications before and after integration (**Figure 5c**). This classification measures how closely an extracted annotation corresponds to the information present in the original text. Integration converted many previously exact matches into semantic matches or no matches. This pattern is consistent with METI replacing text-supported LLM annotations with values that were less specific, expressed at a different level of granularity, or not directly supported by the publication text.

Overall, METI integration substantially increased metadata coverage, particularly for information that is difficult or impossible to recover from publication text alone, such as PTMs and mass analyzer information. Nevertheless, this additional coverage came at the cost of reduced granularity and lower agreement with the SDRF-Proteomics reference for several fields. These results demonstrate that repository-derived metadata should not automatically be treated as authoritative when it conflicts with independently extracted annotations.

The 127 cases in which HAMLET populated a field left unannotated in the SDRF-Proteomics reference could not be scored as correct or incorrect against that reference, since no gold-standard value existed for comparison. The LLM-as-a-judge framework resolves this by evaluating each extracted value directly against the manuscript text rather than against SDRF, which also makes it possible to hold the SDRF-Proteomics reference itself to the same text-grounded standard applied to HAMLET, rather than treating it uncritically as ground truth. We compared three sources under this framework across the same 30 papers used throughout this evaluation: HAMLET's raw, pre-harmonization per-agent output; HAMLET's harmonized output after integration with repository- and raw-data-derived evidence; and the expert-curated SDRF-Proteomics record.

Judged directly against the manuscript text under the strict evaluation standard, accuracy was highest for HAMLET raw (85.2%), intermediate for HAMLET harmonized (73.1%), and lowest for the SDRF-Proteomics reference (54.1%) (**Figure 5d**). Extending credit to values a domain expert would confidently infer from context, rather than only values stated or defaulted explicitly, raised each source's accuracy only modestly (HAMLET raw: 87.5%; HAMLET harmonized: 75.5%; SDRF-Proteomics: 57.1%). This small strict-to-inference gap indicates that HAMLET's inference behavior is already well calibrated rather than overly conservative:

most of what a permissive, expert-level standard would additionally credit, HAMLET already extracts and marks explicitly, so relaxing the standard closes little of the remaining gap to full accuracy. Consistent with this, a share of the values HAMLET raw's strict-mode source check flagged as hallucinated were reclassified as legitimate inference once the relaxed standard was applied, with the hallucinated share falling from 3.3% to 2.0% (**Figure 5e,f**) some of what looked like fabrication under the stricter test was in fact defensible inference the strict standard was simply not designed to credit.

Harmonization changes the error profile in a perhaps unexpected manner. The correct share falls by 12 percentage points from raw to harmonized output (85.1% to 73.0%). This drop does not reflect new fabrication, since the hallucinated share stays essentially flat between the two (3.3% to 3.1%). The change happens by three distinct mechanisms.

First, harmonization instead adds a technical-origin category absent from the raw output entirely (0% to 6.7%), crediting values recovered from METI's raw-data pipeline that the manuscript never states, and it enlarges the type-mismatch category (2.8% to 6.7%). This increase concentrates almost entirely in two fields that harmonization actively rewrites, experimental design (0% to 67% type mismatch) and organ (0% to 26%), while every other field's error profile stays close to unchanged between raw and harmonized output (Figure 5H-J).

Second, the harmonization reduces the possibility of verbatim text matches due to ontology normalization. This is not a hallucination but type mismatches and wrong values. Example of this is clear in the organ category where the raw extraction reports breast but the normalized extraction becomes breast cancer cell. Because the organ field definition excludes cell-type and material-type language, a normalized value of this kind fails the type check despite remaining biologically accurate and more specific than the value it replaced.

Third, harmonization occasionally trusts a technical or repository cross-reference over the manuscript wording. This mechanism is exactly what the distinction between `judge_accuracy` and `judge_accuracy_adjusted` was designed to separate: a value can match the deposited data while diverging from what the paper itself reports. Harmonization therefore trades a text-grounded value for one that is either more specific than the organ field allows or drawn from a source that outranks the manuscript, in a small number of fields, echoing the granularity cost already visible in the SDRF-referenced integration analysis above (Figure 5A-B).

Judging the SDRF-Proteomics reference by the same text-grounded standard reframes what expert-curated means for a subset of fields. SDRF-Proteomics' lower overall accuracy comes almost entirely from hallucination (37.0% of all judged annotations), while type and value errors occur at rates comparable to the two HAMLET sources. Within the judge framework, a hallucinated verdict means only that the value cannot be connected to the manuscript text; it carries no claim that the original curator invented the value or recorded it incorrectly. This hallucinated share concentrates in a small number of fields, overwhelmingly the same experimental design metadata already identified as hardest to recover in the human-annotation and text-mention analyses above (Figure 1C-D): factor value (83% hallucinated), fractions (81%), technical replicates (72%), and age (66%) are the four highest single-field hallucination rates observed anywhere in this comparison (Figure 5H-J). The expert curators most often supplied a value for exactly the fields that neither the manuscript text nor a text-

grounded judge can verify, consistent with these values being drawn from full-text sections, supplementary material, or the curator's direct knowledge of the study rather than from the abstract and Methods text that HAMLET and the judge both use. Direct author knowledge is precisely the resource that a post-hoc extraction pipeline, human or automated, does not have access to, which is the same limitation [REF lesSDRF²³] reports for experimental design metadata specifically. Together with the inter-annotator agreement and text-mention results in Figure 1, this strengthens the case for capturing SDRF at the point of submission, when the original authors still hold the information that no downstream reconstruction, however careful, can fully recover.

This pattern is unlikely to be an artifact of judge inconsistency. Running the complete judging pipeline three independent times returned identical verdicts for 94.4% of annotations and identical fine-grained error categories for 90.0%, with per-paper accuracy varying by only 2.2 to 3.7 percentage points in standard deviation across sources and modes (Figure 5G). Run-to-run disagreement was, as expected, higher in inference mode than strict mode for every source, since deciding whether a value is confidently inferable is a more subjective call than checking for an explicit textual match. One field-level exception is worth flagging for caution: material type showed the highest inter-run disagreement observed for any field in this comparison (28%, in the SDRF-Proteomics source), so its accuracy figures should be treated as less certain than the other high-hallucination fields identified above.

Taken together with the integration analysis above, these results narrow where HAMLET's remaining errors concentrate. Harmonization's accuracy cost traces to a specific, identifiable trade in a small number of fields, where repository-derived or ontology-normalized evidence overwrites a more specific, text-supported value, most consequentially for organism identity and experimental design structure. HAMLET's own inference behavior is already conservative enough that a more permissive standard adds little additional credit, so the remaining gap to full accuracy reflects genuine extraction limits rather than an overly strict evaluation. Experimental design remains the hardest category to ground in text for every annotator in this comparison, human and automated alike, consistent with every other analysis in this paper that has examined these fields directly.

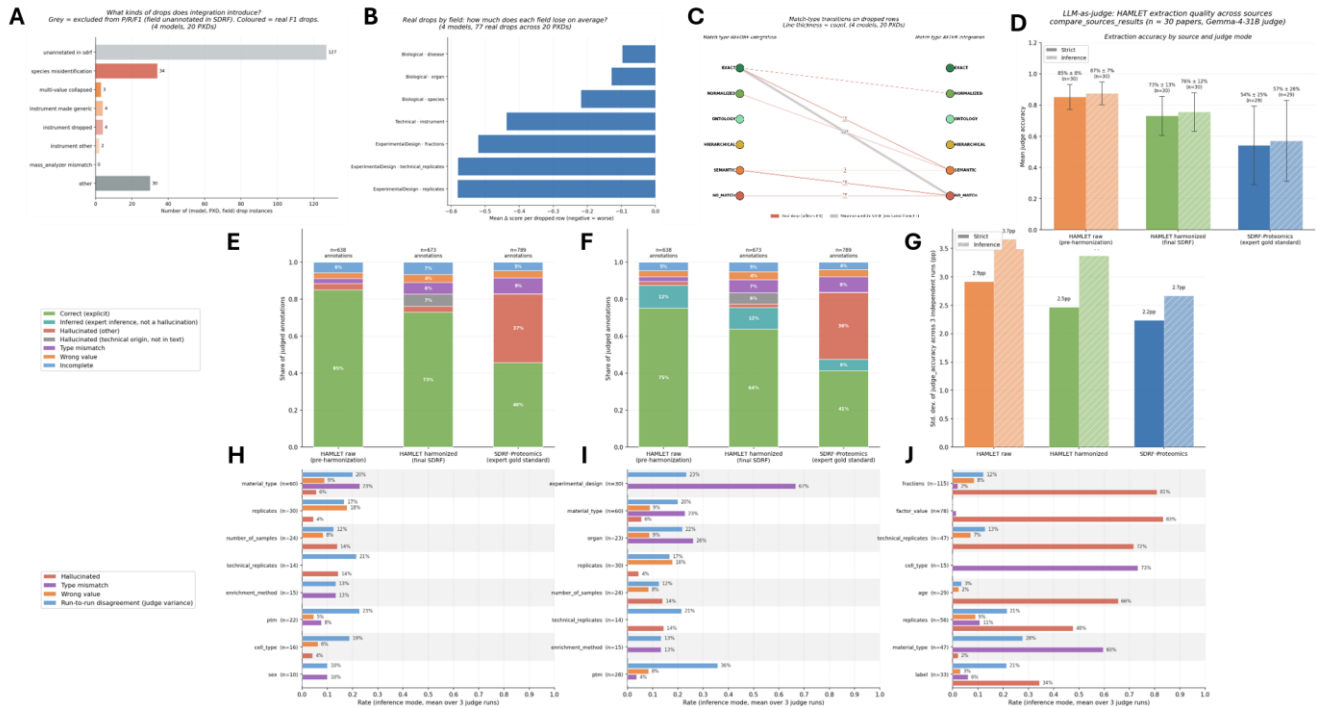


Figure 5. Integration and LLM-as-a-judge evaluation of HAMLET's harmonized output against SDRF-Proteomics and the manuscript text.

(A) Categorization of score changes introduced by integration across 4 models and 20 PXDs (204 total model-PXD-field instances). Grey bars mark fields left unannotated in the SDRF-Proteomics reference and excluded from precision, recall, and F1; colored bars mark the 77 real, scoring-relevant drops. (B) Mean change in match score for the 77 real drops, by field (4 models, 20 PXDs). (C) Match-type transitions for dropped rows, before versus after integration (4 models, 20 PXDs); line thickness indicates count. (D) Mean per-paper judge accuracy by source and evaluation standard (Gemma-4-31B judge, compare_source_results), error bars show standard deviation across $n = 30$ papers for HAMLET raw and HAMLET harmonized and $n = 29$ for SDRF-Proteomics. (E) Error composition under the strict evaluation standard for HAMLET raw ($n = 638$), HAMLET harmonized ($n = 673$), and SDRF-Proteomics ($n = 789$), pooled across all judged annotations. (F) Error composition under the inference-credited standard for the same three sources. (G) Standard deviation of per-paper judge accuracy across three independent judging runs, by source and evaluation standard. (H–J) Per-field hallucination, type-mismatch, wrong-value, and run-to-run disagreement rates (inference mode, mean over three judge runs, fields with ≥ 10 judged rows) for HAMLET raw (H), HAMLET harmonized (I), and SDRF-Proteomics (J).

HAMLET generates a metadata-enriched public proteomics resource at repository scale

Before scaling the framework to the full repository, we first validated the complete pipeline at benchmark scale, applying it to the same $n = 299$ PXD datasets used for the human-annotation comparison above (**Figure 1b**) and comparing the result against existing PRIDE metadata (**Figure 6a**). HAMLET closed most of the gap between deposited metadata and the information actually available for a dataset. Existing PRIDE metadata already saturated two

fields, species and instrument, for which HAMLET's own coverage sat marginally lower (both -4 percentage points, pp), consistent with its policy of returning a field as unknown rather than asserting a value it cannot support with evidence. Every other field improved, and the improvement was largest precisely where the existing record was thinnest: technical acquisition parameters nearly absent from the existing record, including mass analyzer (+95 pp) and fragmentation method (+92 pp), and every experimental design field, including all replicate and fraction counts and factor value (+94 to +96 pp), moved from near-total absence to near-complete coverage. Biological descriptors occupied an intermediate band, improving from 13 pp for tissue to 63 pp for cell type. Only one field showed no gain at all: enrichment method (+0 pp), for which existing PRIDE metadata and HAMLET were equally uninformative. This comparison measures presence, not correctness: a field can appear populated, by PRIDE, by HAMLET, or both, yet still misdescribe the sample, for example when a record reports tissue as liver although the material is in fact a liver-derived cell line.

Not all of this gain originates in the manuscript text. Plotting each field's improvement against the fraction of the 27 gold-standard papers that mention it at all sets an approximate ceiling on what mining the text alone could ever recover, and separates fields whose gain tracks that ceiling from fields whose gain exceeds it (**Figure 6b**). Most biological fields, including tissue, disease state, cell line, and cell type, fall close to the diagonal that marks where improvement equals text-mention prevalence, indicating that their recovery is bounded by, and scales with, how often the field is stated in the text. Species falls well below the diagonal despite being mentioned in nearly every paper, because existing PRIDE metadata already saturates the field and leaves little room for HAMLET to add. Several technical fields, most clearly mass analyzer and other instrument-derived parameters, sit well above the diagonal: their improvement exceeds what text mentions alone would predict, because this coverage draws on repository metadata and on raw-data-derived technical inference (see the technical pipeline validation above) rather than on the manuscript. Every experimental design field sits above the diagonal for the same reason in a different guise, since these fields are more often inferred from a description of the experimental design than stated as an explicit fact, so their recovery again outpaces simple text-mention prevalence. This benchmark-scale result is consistent with the human-annotation comparison established earlier: where GPT's disagreement with the consensus stayed within the range of disagreement between independent human annotators (**Figure 1b**), the combined pipeline recovers metadata at a rate that exceeds what text mining alone would predict, precisely where raw-data-derived or repository-derived evidence is available to supplement the manuscript.

Having established that the complete pipeline recovers metadata reliably at benchmark scale, we then applied it to the full repository.

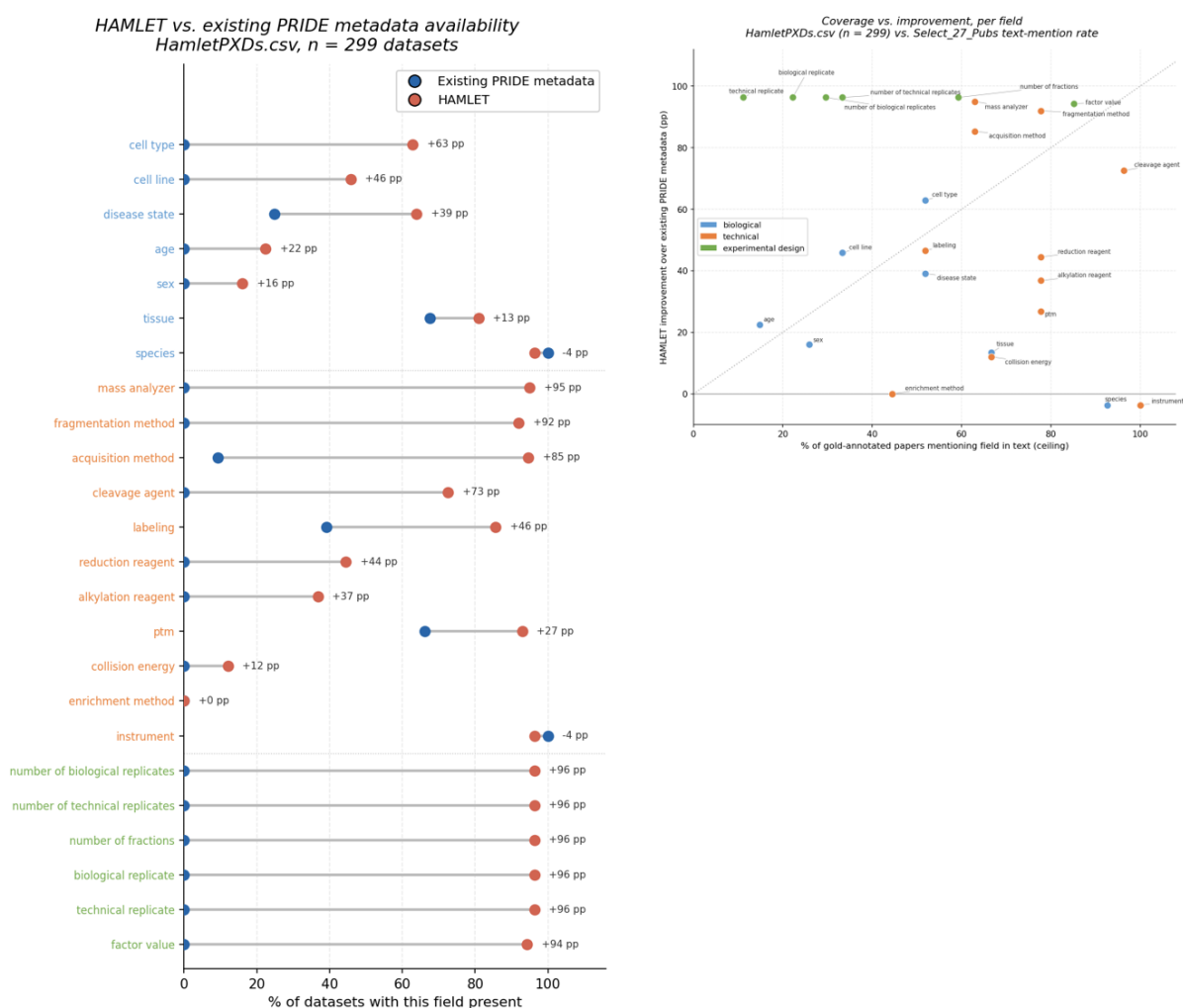


Figure 6. Benchmark-scale validation of the complete HAMLET pipeline against existing PRIDE metadata and the manuscript text-mention ceiling.

(a) HAMLET availability versus existing PRIDE metadata across $n = 299$ PXD datasets (HamletPXD.csv), by field, grouped by biological, technical, and experimental design class; labels show the absolute percentage-point change.

(b) HAMLET improvement over existing PRIDE metadata (y-axis) plotted against the percentage of the 27 gold-annotated papers mentioning each field in text (x-axis, Select_27_Pubs), by field. The dashed diagonal marks where improvement equals text-mention prevalence; points above the diagonal indicate recovery from structured PRIDE data or raw-data-derived inference rather than from text mining alone.

We then applied the complete framework at repository scale to generate a metadata-enriched public proteomics resource. In total, HAMLET processed 15,000 PRIDE datasets with associated open-access publications, corresponding to [N_raw_files] raw MS files and [N_publications] full-text articles (Figure 5a). Across these datasets, the pipeline extracted [N_values] metadata values spanning biological context, experimental design, and technical acquisition or sample preparation properties. Of these, [N_normalized] values were normalized to controlled vocabulary or ontology identifiers, enabling comparison across datasets despite variation in manuscript terminology, repository descriptions, and submitted annotations.

The resulting resource substantially increased metadata availability across multiple dimensions of dataset reuse. Biological fields, including species, tissue, cell type, cell line, disease state, and sample source, were recovered for [x%] of datasets. Technical fields, including instrument platform, acquisition mode, labeling strategy, digestion enzyme, enrichment method, fractionation, and modification-related metadata, were recovered for [x%] of datasets. Experimental design fields remained the most challenging, but HAMLET nevertheless recovered case/control status, treatment or perturbation factors, replicate information, sample counts, or fractionation structure for [x%] of datasets.

The resource is released as a versioned metadata corpus through GitHub at [GitHub repository] and archived through Zenodo, as well as integrated into ProteomeXchange through a searchable interface for community use. Users can query projects by biological attributes, experimental design features, technical acquisition parameters, and ontology identifiers. Each query result links back to the original ProteomeXchange accession, associated publication, and extracted evidence. This makes the resource usable both as a discovery layer for researchers seeking reusable datasets and as a curation layer for identifying incomplete, conflicting, or high-value metadata records.

Metadata enrichment enables biologically and technically meaningful dataset discovery

To show that the enriched metadata layer supports repository-scale data discovery, we evaluated HAMLET on three use cases that represent common modes of proteomics data reuse: biological reanalysis, selection of training data for artificial intelligence models, and technical benchmarking across acquisition strategies.

The first use case was biological reanalysis. Case/control structure is among the most valuable fields for reuse but often the hardest to recover since it is rarely recorded in a structured repository metadata and instead has to be recovered from the manuscript.

Querying for human case/control disease proteomics datasets in a specified tissue returned [N_case_control] projects, including datasets annotated with normalized disease, tissue, species, and experimental comparison terms (**Figure 5b**).

An equivalent keyword search over PRIDE returns [Nkeywordbio] datasets of which [Nrelevantdatasets] carry a recoverable case/control comparison.

These included [example disease area 1], [example disease area 2], and [example disease area 3] datasets with evidence-supported case/control or disease/control labels.

The second use case was the selection of training data for artificial intelligence models. The difficulty here is not finding matching datasets but excluding confounded ones, because the disqualifying property, targeted enrichment or PTM-focused sample preparation, is never stated as structured metadata and must be inferred jointly from the manuscript and the spectra. We queried for human tissue datasets acquired by DDA on Orbitrap instruments under label-free quantification, then removed every dataset with manuscript or spectral evidence of enrichment or PTM-focused design (**Figure 5c**). The positive criteria alone returned [N_positive] datasets; the negative criterion removed [N_excluded] of them, leaving

[N_clean] suitable for training, a distinction a faceted keyword search cannot make because it can only filter on what is present, not on what must be absent.

The third use case was technical benchmarking across acquisition and sample preparation strategies, revealing unique properties of the HAMLET corpus. A query for phosphoproteomics enrichment datasets identified [N_phospho] datasets with evidence for phosphopeptide enrichment, phosphorylation-related modification signals, or manuscript-supported phosphoproteomics study design (**Figure 5d**). These datasets spanned [N_instruments] instrument models, [N_labeling_modes] labeling or quantification strategies, and [N_species] species. Because HAMLET separates observed modification evidence from enrichment or study-design evidence, users can distinguish datasets in which phosphorylation is detected during open search from datasets designed specifically for phosphoproteomics. This distinction is important for benchmarking enrichment-aware workflows, modification localization tools, open-search algorithms, and repository-scale reanalysis pipelines.

Across the three use cases, HAMLET changed dataset discovery from keyword-based browsing to structured, evidence-aware retrieval. Queries could combine biological terms, technical parameters, experimental design variables, ontology identifiers, and confidence filters, while retaining links to the supporting manuscript text, raw data-derived evidence, or PRIDE record. This allows users to inspect why a dataset was returned and to decide whether the metadata evidence is sufficient for their intended analysis. Together, these results show that HAMLET converts disconnected public proteomics datasets into a searchable metadata layer for systematic reuse.

TODO: create figure

Figure 5. HAMLET generates a metadata-enriched public proteomics resource at repository scale.

(a) Repository-scale processing summary showing the number of PRIDE/ProteomeXchange datasets, associated open-access publications, raw mass spectrometry files, and successfully extracted metadata entities. [IAN]

(a) Biological reanalysis query for human case/control disease proteomics datasets. [IAN]

(c) Artificial intelligence training-data query for human tissue datasets acquired by DDA on Orbitrap instruments using label-free quantification, without targeted enrichment or specific PTM-focused sample preparation. [IAN]

(d) Query for phosphoproteomics enrichment benchmarking, summarized by instrument platform, labeling strategy, and species. [IAN]

Discussion

We developed and validated HAMLET, a hybrid framework that reconstructs the biological, experimental, and technical context of public MS-based proteomics datasets by integrating publications, raw data, repository records, and ontologies. The performance we report matters less than how it was obtained. The clearest evidence for this is the model comparison: Gemma 4 31B, an open-weights model, reached an overall extraction F1 on par with GPT-5.4 and ahead of Claude Opus 4.5, Gemini 3.5 Flash, and Qwen 3.7 Max once placed inside the HAMLET framework, while the same framework's absence collapsed output volume into an unpredictable, thirteen-fold difference between models the moment the specialized agents were replaced with a single general prompt. The model was the variable that changed least; the framework was what carried the result, both in accuracy and in whether the output could be trusted to look the same twice. HAMLET constrains the language model at every step with structure it cannot invent, grounding each extracted value in controlled vocabularies and ontologies, pairing it with the evidence that supports it, and checking it against the other available sources. An ungrounded model left to read freely is unreliable because it hallucinates, and it is unusable at scale because its output does not generalize, since free text that is correct for one dataset cannot be compared across datasets, and comparison across datasets is the entire purpose of machine-readable metadata. The ontology is what turns a plausible string into a value that can be queried, aggregated, and reasoned over. We emphasize this because intuition runs the other way. As models grow more capable, it is tempting to expect that standards and ontologies will matter less, and that a sufficiently capable model will simply understand what a study did. Our results point the opposite direction, that the more we rely on models to read the literature at scale, the more their output has to be anchored to shared structure. Indeed, ontologies and community standards have often been treated as bureaucratic overhead, but in the age of large-scale automated extraction they are the precondition for that extraction to mean anything.

This grounding is what makes the resource queryable in ways the repository alone does not yet support. The use cases we present are not only demonstrations of recall, because they are queries that could not be run against the existing state of metadata, given that the fields they depend on were never structured, were stated only in the manuscript, or were present in the raw data while absent from the annotation. Selecting datasets for model training by excluding those with evidence of enrichment or separating studies designed for phosphoproteomics from those in which phosphorylation merely appears during open search, requires evidence that no single source holds alone. This reframes metadata from a static label attached at deposition to a first-class scientific output in its own right, one that can be incomplete when a dataset is deposited and improved later as extraction and validation methods advance. It can be reconstructed, corrected, and enriched long afterwards, facilitated by a resource like HAMLET.

That reconstruction depends entirely on the ability to read the manuscript, which makes the current state of open access untenable. A dataset deposited in PRIDE under an open license, but described only in a manuscript behind a paywall, is not fully open in any sense that matters for reuse. The practical implication is that the interpretive context a dataset needs, must travel with the deposit itself. This can be through richer structured metadata at submission and through deposition of the manuscript or its Methods wherever licensing permits. This also

makes clear why closed access is a problem that better metadata alone cannot solve: a structured record captures the values a schema anticipates, while a great deal of what makes a dataset reusable, the reasoning behind a design, the caveats, the incidental observations, lives only in the prose. The community has rightly built strong norms around data sharing through ProteomeXchange, and the same expectation should extend to the context that gives the data its meaning, in whatever form the publishing system allows researchers to provide it.

Within this landscape, HAMLET is intended to contribute to the existing SDRF-Proteomics efforts and curation workflows. HAMLET addresses the need for metadata annotation of the large body of datasets deposited before the standard was adopted, together with the many recent records that remain incomplete or inconsistent. For historical datasets, it can backfill missing metadata and surface information that would otherwise require manual inspection. For future submissions, the same approach could pre-populate fields for a submitter to verify, reducing annotation burden while preserving human oversight. For curators, source-aware conflict labels and confidence scores can prioritize the records most in need of attention. Judged against the same text-grounded standard applied to HAMLET, the expert-curated SDRF-Proteomics reference itself could not be connected to the manuscript text for 37% of its annotations, concentrated almost entirely in experimental design fields such as factor value, fractions, and technical replicates. This is not evidence that the curators recorded these values incorrectly; it is evidence that the values came from knowledge the curator held directly rather than from anything a post-hoc reconstruction, human or automated, can recover, the same limitation previously reported [REF reports for experimental design metadata specifically. In each of these roles, automated extraction is a triage and acceleration mechanism, not a substitute for community standards or expert judgement.

That distinction will only grow more important. Automated, LLM-driven metadata extraction is going to become common, and much of it will be built without the safeguards described here. The temptation will be to run a capable model over literature, write its output into structured fields, and treat the result as curated metadata. The danger in that is subtle, because the output looks identical to human annotation once it sits in a database field, and its origin and reliability are no longer visible to anyone who later reuses it. A confidently extracted but wrong value is more harmful than a missing one, because it is acted upon rather than investigated. HAMLET's own integration step makes the case concretely: prioritizing raw-data-derived evidence over the language model's output improved coverage overall, but for a minority of datasets it replaced a correct, text-supported value with an incorrect or less specific one, most often the inferred organism. A closer look at which values changed shows this is not one failure mode but several. Species misidentification, a genuine error in the raw-data pipeline's own taxonomic call, is the largest single contributor. A different and more subtle case appears in the organ field, where two things that are not errors at all can still register as one: ontology normalization occasionally returns a value more specific than the field definition allows (breast becomes breast cancer cell), and harmonization occasionally trusts a technical or repository cross-reference over the manuscript's own wording. Both remain defensible values, just not the value the manuscript reports or the value the field schema was built to hold. Detecting any of this had happened at all depended on the source-aware labels recording which evidence stream produced each value; without them, a genuine misidentification and a defensible reattribution would have been indistinguishable from each other, and from the improvements sitting next to them. Our position is that machine-extracted metadata should carry less

authority than expert human annotation, and that this difference must be encoded in the record rather than left to convention. Provenance, confidence, and the human or machine origin of every value should remain attached to it, distinguishable at the point of reuse, and weighted accordingly. A value flagged as automatically extracted with medium confidence invites the scrutiny it deserves, while the same value stripped of that label silently becomes ground truth. The field is about to receive automatically generated annotations at a scale that will make their origin impossible to reconstruct after the fact, so the distinction has to be designed in now, while the provenance is still known. HAMLET is one model for how to do this, and we expect it will not be the only one, but the principle should outlast any particular implementation.

Several limitations remain, and each points to a direction for further work. First, publication mining is bounded by access and licensing, so HAMLET can process only full-text articles available for text mining, and metadata locked in closed-access manuscripts or restricted supplementary formats are missed. This is the practical face of the open-access argument made above, and its remedy is the same. Second, manuscripts frequently omit information essential for reuse, particularly replicate structure, sample-to-file mappings, and fractionation schemes. Where information is absent or ambiguous, HAMLET reports unknown or lower-confidence values and avoids forcing a misleading annotation, which leaves honest gaps that richer capture at submission time, as argued above, would fill.

Third, experimental design is the category most sensitive to the extraction framework and the hardest to synthesize from narrative text, tables, figures, and supplementary materials, even though a sufficiently capable model inside the complete HAMLET framework recovers it about as reliably as the other classes; this is also where expert human annotators agreed least, so the residual difficulty is intrinsic to the task itself. The same fields are also where the deposited SDRF-Proteomics reference itself is least connected to the manuscript text, judged by the identical standard applied to HAMLET, which indicates that the difficulty is not merely disagreement between annotators but a genuine absence of recoverable textual evidence. A value that humans cannot consistently assign is one that should be flagged for review rather than trusted from any automated source. Fourth, the source-aware priority given to raw-data-derived evidence over the language model's output is a reasonable default, not a guarantee: for a minority of datasets it degrades an already-correct, text-supported value, so the priority rule itself needs the same conflict labeling and confidence scoring applied to every other value, rather than being treated as unconditional ground truth. Finally, even with evidence constraints, ontology grounding, and validation, automated metadata cannot fully replace expert review in high-stakes analyses, so the resource is best used as an evidence-linked discovery and prioritisation layer, with manual verification retained wherever dataset choice materially affects a biological conclusion.

Future development should focus on deeper integration with ProteomeXchange and PRIDE and on iterative improvement through curator feedback. Coupling extraction with submission tools would let authors verify suggested annotations before deposition, which moves enrichment upstream of the gap and avoids perpetually chasing it after the fact. Extending the framework beyond bottom-up discovery proteomics to specialised modalities, including crosslinking mass spectrometry, hydrogen-deuterium exchange, and metaproteomics, will require field-specific schemas and validation strategies. The same evidence-integration principles should translate to metabolomics and other omics domains, where context is

similarly scattered across raw files, publications, repository records, and controlled vocabularies. Most importantly, a closed feedback loop, in which curator corrections, user feedback, and adjudicated conflicts are returned to the extraction prompts, normalisation rules, and benchmark corpus, would let the resource improve across successive releases, which is what treating metadata as a first-class, improvable output requires in practice.

By making the biological and technical context of public proteomics datasets computable, HAMLET provides infrastructure for dataset discovery, artificial intelligence development, repository curation, and large-scale biological reanalysis. The wider lesson reaches past proteomics. As automated extraction becomes the default way that public data is described, the systems we build to read that data will only be as trustworthy as the structure we ground them in, and the standards and ontologies that much of biology has treated as housekeeping turn out to be what makes large-scale reuse possible at all.

Materials and Methods

NLP pipeline

Manuscript selection and extraction for training purposes

Two independent manuscript corpora were assembled for training and evaluation of the NLP pipeline. The first corpus was built around the 2025 EuBIC-MS hackathon effort. Candidate manuscripts were obtained from PRIDE, selecting projects that had a corresponding SDRF file alongside the raw data, in the bigbio GitHub repository. To ensure sufficient diversity of experimental designs and to minimise bias in training examples, the corpus was curated to span a range of quantification strategies and acquisition methods, including label-free quantification (LFQ), SILAC, TMT, data-dependent acquisition (DDA), and data-independent acquisition (DIA).

The second corpus was assembled by collaborators at NCEMS (National Synthesis Center for Emergence in the Molecular and Cellular Sciences), who curated an SQL database of open-access proteomics publications linked to PRIDE accessions (PXD identifiers), totalling XXX entries from which XXX were selected for manual annotation.

For each manuscript, the license was checked to ensure text mining was permitted under the copyright terms. Then abstract and material and methods were extracted as plain text.

Gold standard annotations

EuBIC

The primary annotation set was produced across two phases: a prep-phase prior to the EuBIC-MS Developers Meeting, and an annotation phase during the hackathon itself. In the preparatory phase, a core team drafted initial annotation guidelines and performed two iterative annotation rounds, each covering approximately 20 documents. After each round, inter-annotator agreement was assessed, results were discussed, and the guidelines were refined accordingly. During the hackathon, two additional annotation rounds were completed collaboratively, bringing the total manually curated corpus to approximately 75 papers covering around 60 metadata entities. All annotations were performed using the BRAT annotation tool. Furthermore, following the hackathon, the full corpus was re-annotated independently by a single annotator, providing a complete single-annotator reference set across all documents. This allowed direct comparison between the multi-annotator consensus and a single-annotator baseline, enabling assessment of how annotation strategy affects ground truth quality and downstream model performance.

NCEMS

From the NCEMS corpus and additional 20 publications were manually annotated by a single expert annotator. These annotations followed the same metadata fields as the EuBIC annotation schema, enabling direct comparison across the two corpora.

BigBio

Consensus annotations

To derive a single ground truth annotation per manuscript, all individual annotations were combined and reduced to a consensus in two steps. First, for each metadata field, annotators with excessive disagreement relative to the group, aka mis-annotations, were identified and

excluded. Among the remaining annotations, the consensus term for each field was determined by majority agreement. Second, the level of support for each entity type was assessed by counting how many annotators had independently assigned it: entity types mentioned by only a single annotator were excluded from the consensus, as isolated annotations were considered insufficiently reliable to serve as ground truth.

Inter-annotator agreement

Pairwise agreement between human annotators was quantified using Cohen's kappa per entity. To accommodate for lexical variability, semantic similarity between annotations was assessed using SciBERT-derived cosine similarity.

Model selection

Five LLMs were evaluated for the extraction pipeline: GPT-5.2, Gemini 2.5, Llama 4 Scout, DeepSeek R1 and Claude Opus 4.6. These were selected to represent a range of architectures and proprietary API-based models and locally deployable alternatives.

Temperature optimization based on annotation consistency for Llama and DeepSeek

For Llama and DeepSeek, temperature was optimized prior to evaluation. A range of temperature values were systematically tested and manually checked for extraction consistency. A temperature of 0 was selected for both models to obtain deterministic and reproducible outputs. For the other models, default temperature settings were used.

Prompt optimization

Prompt design was developed iteratively. In the first stage, a single generalized prompt, the so-called master prompt was used to extract all metadata simultaneously. In the second stage, this prompt was decomposed into three dedicated prompts for (i) biological characteristics; (ii) technical parameters and; (iii) experimental design. This improved granularity.

Started from main prompt => granularity with dedicated prompts

[Insert section about how we introduced rules and examples to the prompts which improved performance]

Technical extraction pipeline

Workflow implementation and HPC execution

The workflow was implemented in Nextflow³¹ (DSL2) as a modular, process-oriented pipeline in which each analytical stage (data acquisition/standardization, run-level parameter inference, peptide/organism inference, PTM discovery, and aggregation) is encapsulated as an independent process with explicit input/output file contracts. This design enables robust execution across heterogeneous datasets, simplifies provenance tracking by preserving intermediate artifacts at each stage, and supports both single-study execution and high-throughput batch processing across multiple ProteomeXchange/PRIDE accessions.

Pipeline execution on the high-performance computing (HPC) systems was orchestrated by Nextflow using containers where appropriate to maintain software viability, longevity, and version control. Nextflow's built-in caching was used to avoid wasted computation when re-

running analyses with unchanged inputs, while still allowing selective re-execution of downstream steps when upstream inputs or parameters change. The workflow also adopts a “best-effort metadata” philosophy for non-critical stages (where appropriate), allowing the pipeline to complete and emit partial metadata products even if a subset of optional enrichment steps fail, while preserving logs and per-process outputs to support downstream auditing and troubleshooting.

Fetching and standardizing raw proteomics data

PRIDE studies were provided to the workflow either as a single accession (PXD) or as a CSV list of PXDs for batch execution. For each PXD, the workflow first queried the PRIDE Archive web service to retrieve project-level metadata and enumerate all publicly available file records. The complete response (including project metadata and file listings) was preserved as a JSON provenance artifact for each study, enabling traceability between downstream metadata products and the original PRIDE records used to drive data acquisition.

To standardize raw instrument data into a consistent downstream-compatible representation, the workflow selectively downloaded Thermo .raw files from PRIDE public file locations (preferring HTTPS when available, with fallback to FTP) using a resilient download strategy with multiple retries and exponential backoff to tolerate transient network or server errors. For high-throughput or bandwidth-constrained environments, downloads optionally used multi-connection segmented transfer (aria2c), and the number of raw files per study could be capped to control runtime and storage footprint during exploratory or large-scale runs. Downloaded .raw files were converted to mzML using ThermoRawFileParser (configured via a container-provided binary and/or environment variable), producing analysis-ready mzML files in the same per-study directory structure. Conversion was performed on a per-file basis to maximize robustness: failures to convert individual files were logged and processing continued where possible, while studies with no downloadable raw files were flagged and skipped to avoid generating misleading downstream metadata from empty inputs.

Run Assessor-based acquisition and labeling inference

To minimize manual configuration and to support heterogeneous PRIDE submissions, run-level acquisition and labeling properties were inferred directly from the standardized mzML files using an automated “Run Assessor” step. After raw-file conversion, the workflow executed a run assessor routine on each study directory to extract instrument- and run-level descriptors from mzML metadata and scan content. The assessor summarizes per-file spectral statistics (e.g., MS level counts, fragmentation characteristics, and precursor/isolation information) and produces a study-level JSON report that serves as the primary evidence source for downstream auto-configuration.

Acquisition mode (DDA vs DIA) was inferred using precursor isolation window characteristics derived from MS2 spectra. In brief, the pipeline examines the distribution of observed isolation window widths and classifies runs as DDA when windows are narrow (typical of precursor-targeted fragmentation) and as DIA when windows are wide (typical of multiplexed fragmentation), with an “ambiguous/unknown” designation used when the observed widths do not clearly support either regime. In parallel, labeling and quantification mode were inferred from the assessor’s labeling summary, which provides a categorical call (e.g., label-free vs isobaric tags vs metabolic labeling) along with scores reflecting support for alternative labeling hypotheses. The pipeline then converts these calls into standardized downstream

configuration fields (including a labeling category and a derived quantification type) and writes them to a compact “detected parameters” JSON artifact. This artifact is subsequently consumed by downstream modules to (i) parameterize DDA vs DIA processing logic and (ii) select labeling-aware modification settings used in later PTM-focused search steps.

De-novo peptide identification

To generate organism- and modification-relevant evidence without requiring a curated, study-specific search database, the workflow performs de novo peptide identification directly from mzML files. The de novo engine is selected automatically based on the acquisition-mode inference from the Run Assessor step. For data inferred as data-dependent acquisition (DDA), de novo sequencing is performed with Casanovo using GPU acceleration. For data inferred as data-independent acquisition (DIA), sequencing is performed with Cascadia (also GPU-accelerated) using a pre-trained model checkpoint. In both cases, each mzML file is processed independently and produces a per-file set of peptide-spectrum match (PSM)-like identifications with an associated de novo confidence score.

To standardize outputs across engines and enable uniform downstream handling, de novo identifications are normalized into a common tabular format. Casanovo outputs are retained in mzTab-like form and parsed to extract peptide sequences and scores; Cascadia sequencing results (native “.ssl” format) are converted into a mzTab-like table with harmonized columns (sequence, charge, retention time, spectrum reference, and score). Identified peptides are then filtered to enrich for high-confidence evidence suitable for downstream metadata inference: peptides are required to exceed a minimum sequence length (to reduce ambiguity in taxonomic matching) and to meet user-configurable de novo confidence thresholds (typically applied as one or more cutoffs to generate multiple evidence sets). Finally, exact-match contaminant peptides are removed using an in silico digest of a standard contaminant FASTA (tryptic digestion with missed cleavages) prior to any downstream organism- or PTM-oriented interpretation.

Organism inference using Peptonizer

For each raw file converted to mzML, we inferred likely source organism(s) directly from peptide evidence, without requiring an a priori protein database. Based on the acquisition-type inference described above, de novo sequencing was performed using Casanovo for DDA data or Cascadia for DIA data, producing peptide-spectrum matches with an associated confidence score.

De novo identifications were post-processed to generate peptide lists for taxonomic inference. Predicted sequences were “stripped” of modification annotations to obtain an unmodified peptide string, and then filtered for minimum length (>10 amino acids) and minimum confidence score. To assess robustness to confidence filtering, we generated multiple peptide-evidence sets using score thresholds of 0.60, 0.70, and 0.80, writing both a full table and a slim two-column table (peptide, score) for each threshold.

To reduce spurious organism assignments from common laboratory contaminants, we removed peptides that exactly matched an in silico tryptic digest (trypsin, up to two missed cleavages) of a universal contaminant FASTA, retaining only non-contaminant peptide sequences for downstream taxonomic modelling.

Taxonomic assignment was performed with Peptonizer2000, executed as a Snakemake workflow within the pipeline. Peptonizer2000 combines Unipept peptide-to-taxon mappings with a probabilistic graphical model (PepGM) and belief propagation to estimate posterior support for candidate taxa (tool description: [readme.md](#)). For each mzML file and confidence-filtered peptide set, Peptonizer2000 was run only when at least 100 high-quality peptides were available, and the taxonomic search space was restricted to a predefined list of candidate taxonomic identifiers (a comma-separated taxid query). Peptonizer2000 produced a ranked list of taxa per file (`peptonizer_result.csv`; columns include `taxon_name`, `taxon_id`, `score`), and the highest-scoring `taxon_id` was used as the primary `organism_id`-derived taxid in downstream taxid determination and validation logic.

Modification search

To populate modification-related metadata in a data-driven manner (rather than relying on a fixed, study-specific PTM list), DDA datasets were analyzed using a two-pass workflow coupling SAGE database search with PTM-Shepherd summarization. In the first pass (open search), spectra were searched with a wide precursor mass tolerance to enable broad mass-shift discovery while disabling predefined static and variable modifications to preserve an unmodified baseline for downstream mass-shift interpretation. SAGE produced a PSM-level results table, which was reformatted into a PTM-Shepherd-compatible PSM table (including delta-mass values) and analyzed with PTM-Shepherd to generate global modification summaries (e.g., `global.modsummary.tsv` and `global.profile.tsv`).

Candidate PTMs from the open-search PTM-Shepherd summary were then filtered to obtain a conservative set of “validated” PTM classes for downstream modeling. Specifically, entries labeled as unannotated mass shifts or isotopic peak artifacts were excluded, and modifications were required to exceed minimum abundance criteria (by default, at least 50 PSMs and at least 1% of PSMs in the PTM-Shepherd summary). To ensure that the subsequent closed search remained computationally tractable and chemically interpretable, only PTM classes that could be mapped to a residue-level specificity (e.g., phosphorylation on S/T/Y, oxidation on M, deamidation on N/Q, N-terminal acetylation) were converted into a SAGE `variable_mods` specification; negative and extreme mass shifts were excluded.

In the second pass (closed search), SAGE was re-run using an organism-specific UniProt FASTA derived from the per-file taxonomic assignment (taxid) and the validated PTM-derived `variable_mods` list, thereby restricting the search space to a small, empirically supported PTM set. Closed search retained carbamidomethylation on cysteine as a static modification and used tighter mass tolerances appropriate for targeted PTM assignment. To reduce computational burden in closed search, the workflow preferentially used reviewed UniProt entries (with fallback to the full proteome if the reviewed set was unavailable). Depending on the pooling configuration, the closed search was executed either as a pooled analysis across all mzML files in a project or as sequential per-file closed searches followed by aggregation into a single results table; per-file mode was designed to reduce peak memory/stack usage while preserving project-level PTM discovery from the pooled open search.

If PTM-Shepherd did not produce a global modification summary in the open search, or if no PTMs met validation criteria, the workflow conservatively fell back to reporting open-search results only. The primary metadata products of this module were (i) project-level PTM-Shepherd summary tables from the open (and when available, closed) pass, (ii) the machine-readable validated PTM list used to parameterize closed search, and (iii) the closed-search

PSM table used as the main modification-aware identification output for downstream metadata aggregation.

Agentic pipeline

The extraction pipeline was structured in four layers: extraction, integration, normalization and validation (Fig X) within the DocETL framework[REF]. Abstract and Materials and Methods sections were passed to the extraction layer, and the resulting agent outputs propagated through the remaining layers to generate a structured json output.

Biological, technical and experimental design agents

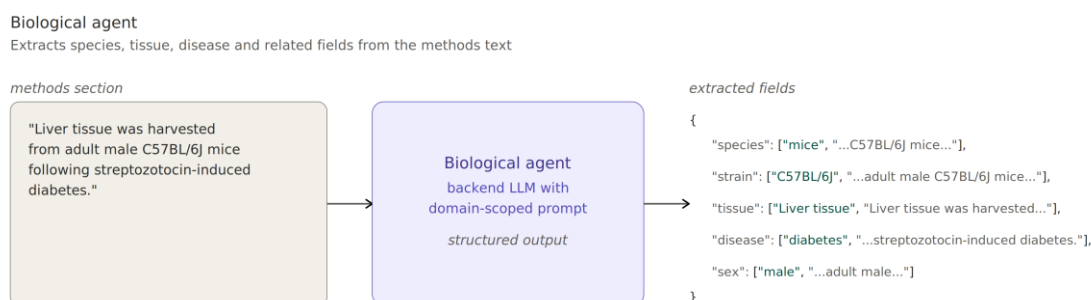


Figure 5. Illustrates biological agent extracting structured json output from unstructured free text

Each input was processed independently by three LLM-based agents, each specialized in a distinct domain of metadata: biological, technical and experimental design. The biological agent extracted sample level characteristics including species, tissue, cell type, cell line, disease state, sample source, age, sex, BMI, strain, and anatomic site of tumor. The technical agent extracted mass spectrometry acquisition parameters including instrument, fragmentation method, ionization type, labeling strategy, cleavage agent, enrichment method, fractionation method, acquisition method, collision energy, reduction reagent, reduction concentration, alkylation reagent, and alkylation concentration. The experimental design agent extracted and inferred study structure information including experimental design type, factor value, number of samples, number of fractions, biological replicate identifiers, technical replicate identifiers, number of biological replicates, and number of technical replicates. A key distinction of the experimental design agent is that its target fields are rarely stated explicitly in the text and therefore require controlled inference from descriptions of experimental procedures. Biological and technical agents, by contrast, extracts values that primarily appear verbatim in the manuscript text.

Chain of thought extraction protocol

All agents were instructed to extract fields systematically and to report each as a two-element list [value, evidence], where the value is the extracted term and the evidence the verbatim sentence that supports it. Fields not mentioned in the text were returned as ["unknown", ""].

This schema was enforced uniformly, ensuring that every valid (non-unknown) value was paired with traceable provenance for downstream verification. Two evidence modes were defined. For verbatim values, the evidence sentence was required to contain the value as an exact substring. For inferred values, the term must be derived from contextual cues rather than stated explicitly, the evidence was prefixed with "inferred: " followed by the verbatim supporting quote. For example, a species value of "Homo sapiens" inferred from a mention of "patient samples" would be reported as ["Homo sapiens", "inferred: patient samples were collected from consenting donors"]. This distinction allowed downstream quality checks to apply appropriate grounding verification to each evidence type.

The biological and technical agents relied primarily on verbatim extraction, with inference restricted to defined cases: (i) species from mentions of patients, donors, or well-characterized cell lines; (ii) tissue origin from cancer type names; (iii) disease state cell line identity; and (iv) labelling strategy as label-free in the absence of any reported labelling. In contrast, the experimental design agent applied inference more broadly, as its target fields including replicate counts, experimental design type, and factor values are rarely stated explicitly and instead require synthesis from descriptions of experimental procedures. Experimental design type was constrained to a controlled vocabulary: treated vs. control, case vs. control, time course, dose response, cross-sectional, or longitudinal. Two default value rules were also applied: number of fractions defaulted to "1" when no pre-fractionation was described, and number of biological replicates defaulted to "1" when none were implied by the text.

Structured validation and retries

DocETL applied structural validation rules to each extraction before accepting the output. For the biological agent, these rules checked that key fields, species, tissue, cell type, and disease state, were returned as two-element lists, and that valid values for species, cell line, cell type, and sample source were accompanied by non-empty evidence strings. For the technical agent, analogous checks were applied to instruments, cleavage agent, labelling strategy, and fragmentation method. For the experimental design agent, validation focused on format and consistency of experimental design type, factor value, number of biological replicates, and number of samples. Extractions that violated any of these rules triggered up to two automatic retries, in which the model was asked to correct the failing fields without regenerating the full output.

Gleaning

Following structural validation, a conditional gleaning step applied up to two rounds of targeted self-consistency review. Gleaning, an LLM-based validation procedure, was triggered only when a valid extracted field lacked supporting evidence. This trigger makes it a corrective step rather than a routine pass. In each round, the LLM re-evaluated its prior output using a domain-specific prompt. This prompt enforced (i) an evidence completeness check; (ii) exact substring matching between extracted value and verbatim evidence sentences; and (iii) replace missing evidence with an "inferred: " prefixed quote. For the experimental design agent, additionally checks enforced (i) exact count values in numeric fields; (ii) collapsing of multi-element lists into a single value; and (iii) validation of inference prefix. A gleaning round returned **"VALID"** if no issues remained.

Ontology Normalization

Following extraction, a normalization agent maps free-text values from all extraction agents to standardized ontology identifiers. Normalization was applied selectively to fields for which a target ontology was defined: species and organism to a curated species subset to a curated subset of 32 commonly studied major model organisms; cell type to the Cell Ontology (CL, ~2,600 terms); cell line to the Cell Line Ontology (CLO, ~40,000 entries); tissue and organ to the Uberon multi-species anatomy ontology (~14,000 terms); disease and disease state to the Mondo Disease Ontology; instrument, labelling strategy, and fractionation method to the PSI-MS mass spectrometry controlled vocabulary; post-translational modifications to Unimod; and sample source to the BRENDA Tissue Ontology (BTO). The following fields did not include ontology mapping: fragmentation method, ionization type, acquisition method, cleavage agent, enrichment method, collision energy, reagent concentrations, and all experimental design fields.

Ontology graph representation

Each ontology was parsed from its source file and represented internally as a directed acyclic graph (DAG). The graph structure ensured two traversal operations for downstream use. Ancestor traversal (up to a configurable depth) was used by the cross-field consistency checker to resolve ontology relationships between co-extracted fields (more below). Lowest-common-ancestor lookup was available for disambiguation between candidate matches when two ontology identifiers disagreed on a field. These hierarchical relationships were used in the benchmark module to enable ontology-aware matching, to ensure a partial correct score for *heart* and *cardiovascular* rather than a complete miss.

The graph maintained two independent lookup structures serving distinct purposes. First, a case-insensitive exact name-to-identifier index, built over all primary names and synonyms at parse time. This supports direct string-based resolution within the cross-field consistency checker without invoking the embedding model. Second, a FAISS-based embedding index (described below) that handles fuzzy matching during normalization. These two mechanisms operated independently and were never interchanged.

Abbreviation handling and synonym injection

Abbreviated binomial names (e.g., *P. falciparum* for *Plasmodium falciparum*) are frequently used in literature but often absent from ontology synonym fields. Two complementary mechanisms were implemented to handle these systematically: automatic synonym injection for the standard binomial abbreviation pattern, and a custom mechanism for registering arbitrary synonyms at runtime.

At index build time, abbreviated forms were injected directly into ontology graph nodes as additional synonyms before embedding. For each non-obsolete node matching the standard binomial pattern, two abbreviated variants were generated and appended to the node's synonym list: a single-letter genus prefix form (*p.falciparum*) and a dot-separated full-genus form (*Plasmodium.falciparum*). Because injection occurred before the embedding step, these forms were encoded as native surface forms of the correct ontology entry and required no special handling at query time. Ontology indices that had already been built and cached before this injection mechanism existed, however, did not contain these forms as embedded synonyms. To cover this case, a query-side expansion rule converted dot-separated full-genus binomials to standard spaced form (e.g., "*Plasmodium.falciparum*" → "*Plasmodium falciparum*") before the embedding lookup; the single-letter genus prefix form is not handled by this fallback, since it is covered by graph injection whenever the index is rebuilt. In all cases,

both the original query and any expanded form were searched independently, and the result with the higher similarity score was returned.

In addition, a custom mechanism to register synonyms during runtime allowed arbitrary variant forms to be associated with specific ontology nodes without rebuilding the full index. Newly registered synonyms were (i) appended to the in-memory graph node, (ii) embedded and appended to the FAISS index, and (iii) persisted to a custom_synonyms.json file in the index cache directory. Persisted synonyms were automatically reloaded and injected into graph nodes on all subsequent pipeline runs, making corrections permanent across sessions.

Embedding-based term matching

Normalisation was performed using SapBERT (*cambridge/tl/SapBERT-from-PubMedBERT-fulltext*), a biomedical language model pre-trained for synonymy-aware entity linking. To reduce computational time, the index was constructed once offline, serialized to disk and reloaded during pipeline execution adding no per-document embedding overhead at inference time. Construction was performed on GPU when available (CUDA) with CPU fallback supported but substantially slower for large ontologies such as CLO.

At build time, all non-obsolete nodes were indexed together with their synonyms. Each surface form, primary name and synonym was embedded independently using the [CLS] token representation of SapBERT (maximum length 128 tokens, batch size 64). This produced a many-to-one mapping in which a single ontology term was represented by multiple embedding vectors, one per surface form, all pointing back to the same ontology identifier. For ontologies containing 10,000 or more indexed surface forms, most notably CLO, an IVF + 8-bit Scalar Quantizer FAISS index was used to balance memory and search latency, with centroids set to min (4096, $N//30$), where N is the number of indexed vectors for that ontology. Smaller ontologies used an exact flat inner-product index.

At query time, extracted values were embedded using the same SapBERT model and matched against the index. The top five candidate ontology terms were retrieved using cosine similarity. This selection allows us to select the highest-ranking candidate for downstream usage and retain the remaining candidates in the output record for manual inspection or downstream reranking if the user would require this. A match was accepted as normalized if the top-ranked similarity exceeded a threshold of 0.70, and was further categorized as high confidence (≥ 0.90) or medium confidence (0.70–0.89). Fields that failed to meet the threshold retained their free-text value with *is_normalized: false* in the output, making normalization failure explicit rather than silently discarded.

Output format

Normalization results were integrated directly into the extraction output. Successfully normalized fields were converted from the raw [value, evidence] format into a structured object containing the original extracted value, the evidence sentence, the matched ontology identifier, the canonical ontology term name, and the similarity score. Unsuccessfully normalized fields that e.g. failed to meet the similarity threshold, retained the [value, evidence] list structure with *is_normalized: false*, ensuring normalization status is always explicit and unambiguous.

Hallucination checks

Following extraction and normalization, three complementary hallucination detectors were applied to identify specific categories of hallucinations: (i) ontological contradiction between co-extracted fields; (ii) negation of the extracted entity within its own evidence sentence; and

(iii) numeric value mismatch between the extracted figure and its quoted source. Each detector produced structured flags targeting a specific failure mode.

Cross ontology consistency

A cross-field consistency checker verified that related biological fields were mutually compatible, using four ontology-based checks. First, cell line–species consistency was assessed using the CLO derives_from relationship, linked to NCBITaxon, and compared against the extracted species using a fuzzy similarity threshold of 0.85. Second, cell line–tissue consistency used the same relationship, mapping BTO identifiers to UBERON. Third, disease–tissue consistency was assessed using the DOID located_in relationship to UBERON. Lastly, cell type–tissue consistency was assessed using the CL part_of relationship to UBERON.

All checks required normalized ontology identifiers, and fields lacking normalization were skipped. Inconsistencies were reported as structured flags with the type of mismatch, the two conflicting field names and values, their ontology identifiers, and the expected value derived from the ontology relationship.

Negation detection

A negation detector identified evidence sentences where extracted values were explicitly contradicted by their evidence (e.g. "TMT" with evidence "No TMT labeling was used in this study."). Negation detection was performed using the NegEx algorithm using the negspacy library on top of a spaCy en_core_web_sm dependency parse. For each field with a non-unknown, verbatim evidence sentence, the evidence was parsed and extracted value was injected as a named entity. NegEX then determined whether the entity fell within the scope of a negation trigger. Fields inherently defined by absence (e.g. label-free) were excluded to avoid false positives, as were inferred evidence cases where negation scope is not directly applicable. Flagged cases were annotated as *negated_evidence* with the corresponding value and evidence sentence.

Numeric mismatch detection

A numeric mismatch detector flagged fields where a numeric value could not be located as an exact substring within its paired evidence quote. The detector recognized a broad set of numeric measurement patterns including molar concentrations (mM, μ M, nM), mass quantities (mg, μ g, ng, mg/ml), collision energies (NCE, eV, kV), mass values (Da, kDa), and temporal and rotational units (min, h, RPM). Matching was strictly exact, as even minor discrepancies constitute factual errors. Fields with inferred evidence prefixes were excluded. Flagged fields were annotated with type *numeric_mismatch* alongside the mismatched value and evidence

Output format

All hallucination flags were aggregated into a `_hallucination_flags` list appended to the output record, written at the same level as the extracted fields to keep provenance, quality metadata, and extracted values co-located in a single record.

LLM-as-a-judge evaluation

Rationale. The hierarchical semantic matching strategy quantifies surface-level agreement between extracted and ground-truth values but cannot distinguish semantically valid extractions that differ in phrasing from genuinely incorrect ones. To complement string- and embedding-based comparison, we implemented an LLM-as-a-judge framework that validates each extracted value directly against the source manuscript text, without reference to any gold-standard annotation.

Evaluation set and extraction sources. We evaluated HAMLET's metadata extraction on a set of 30 published proteomics datasets (PXD accessions), each with a manuscript text (title, abstract, and methods section) and a matching expert-curated SDRF file. For every dataset, three extraction sources were compared against the manuscript text. First, HAMLET raw denotes the unreconciled, per-agent output of HAMLET's three extraction agents (Biological, ExperimentalDesign, Technical), before any cross-agent harmonization. Second, HAMLET harmonized denotes the tool-reconciled output of HAMLET's Integrated Agent, which merges the three agents' extractions and resolves them against structured technical metadata (PRIDE project metadata, PTM-Shepherd output, and RunAssessor search-parameter inference) where available. Third, SDRF-Proteomics denotes the expert-curated ground-truth SDRF for that dataset, serving as the human baseline against which HAMLET is compared rather than merely as a label source. Each source's per-field values were mapped onto a shared set of canonical metadata fields (for example organ, instrument, cleavage agent, and replicates), spanning three categories: Biological, Technical, and ExperimentalDesign metadata.

Judge model and prompting. The judge model, Gemma-4-31B-Instruct (google/gemma-4-31b-it), was accessed via OpenRouter at zero temperature with extended thinking enabled and integrated through the G-Eval framework³² as implemented in DeepEval, which uses chain-of-thought prompting to produce structured assessments through a form-filling paradigm. For each paper, the judge received the title, abstract, and Materials and Methods sections as its sole source of truth, together with the field name, its definition, the candidate value, and the full set of values extracted for that field.

Each extracted value was assessed through four sequential checks. First, a type check verified that the value belonged to the correct metadata category; values of the right category but the wrong specific entity were classified as value errors rather than type mismatches. Second, a source check verified that the value was present in or inferable from the manuscript. A curated set of safe defaults was accepted without explicit textual support, including normal or healthy for disease state in healthy controls, label-free when no labeling was described, developmental-stage terms such as adult when inferable from subject descriptions, and 1 for unreported replicate or fraction counts; values outside both the text and this default set were flagged as hallucinations. For material type and acquisition method, values could additionally be accepted as inferable from the broader experimental context. Third, a truth check verified that the value was factually correct, treating synonyms, abbreviation expansions, and common-to-scientific-name equivalences (for example human and *Homo sapiens*) as correct. Fourth, a completeness check evaluated completeness against the full set of values extracted for that field rather than against the individual candidate in isolation (the sibling-value rule): if the complete extraction set collectively covered the information described in the manuscript, every value within it was marked complete, and a medium verdict was assigned only when

the full set remained genuinely insufficient, for example through subtype drift or partial extraction from a multi-value field. An explicit exception applied to the reduction-concentration and alkylation-concentration fields, for which a numeric quantity paired with a unit (for example 10 mM) was treated as complete in isolation, because the reagent name is captured as a separate entity. Each assessment produced a three-level verdict of high, medium, or low.

Two-pass evaluation. To prevent false extractions from biasing completeness assessment, evaluation proceeded in two passes. In the first pass, all values were judged against the unfiltered sibling list for their field. Values rated low, meaning those flagged as hallucinated, type-mismatched, or factually incorrect, were then pruned from the sibling context, and in the second pass only medium-rated values were re-judged against the cleaned list. This two-pass design ensures that a hallucinated value cannot mislead the judge into concluding that a genuinely correct sibling value is incomplete.

Caching. API calls were minimized using a SHA-256-keyed disk cache indexed on the model identifier, paper identifier, task type, field name, and candidate value, with responses reused across runs. The static system prompt, containing all field definitions, evaluation criteria, and domain rules, was kept separate from the per-paper text to enable prefix caching at the provider level; both the system prompt and the per-paper text were marked for caching, reducing token processing costs for repeated judge calls within the same session.

Strict vs. inference judge mode. For the multi-source comparison described above, each source-field-value triple was judged twice, under two different standards for the source check. Strict mode accepted a value only under the source-check rules described above: explicit textual support, a named safe default, or the narrow material-type/acquisition-method inference allowance. Inference mode extended that same inference allowance to every field, so that a value not literally written in the text but confidently derivable by a domain expert from context stated elsewhere, for example inferring disease from a well-characterized cell line's known origin, or the number of samples by summing explicitly stated per-group replicate counts, was accepted rather than flagged as a hallucination. Values accepted only under this relaxed standard were flagged with an additional inference indicator and tallied separately (`judge_n_inferred`) and were never merged into either the correct or hallucinated counts. The modification field was judged under the strict, explicit-only standard in both modes, since PTM identity cannot be safely inferred. Strict mode thus measures what is explicitly extractable from text alone, while inference mode measures the additional headroom available once expert-level inference is credited.

Correcting for real, non-fabricated technical metadata. A value can be correct without being stated anywhere in the manuscript text, for example an instrument parameter recovered from the raw PRIDE submission or inferred by RunAssessor from the search-engine configuration. To avoid scoring these as fabrications, we built a per-dataset technical reference set from the pipeline's own structured metadata (PRIDE project metadata, PTM-Shepherd output, and RunAssessor-inferred search parameters) and matched each text-judged hallucination against it, using exact match after normalization or token-subset and overlap matching for longer controlled-vocabulary names. Values matching this reference were reclassified out of the hallucinated category into a separate technical-origin category and credited in an adjusted accuracy metric, reported alongside the raw, text-only accuracy.

Error categories and accuracy metrics. In its base form, each annotation was assigned to one of five mutually exclusive outcome classes: correct (type correct, factually valid, complete, and not hallucinated), hallucinated (absent from the manuscript and not a safe default), type mismatch (belonging to a fundamentally different field), wrong value (present in the text but incorrectly applied or factually wrong), and incomplete (correct, but the full extraction set does not cover the complete picture). Per-paper accuracy was computed as the fraction of correct annotations over all predicted annotations and aggregated across the corpus to produce per-paper and per-model performance profiles.

For the multi-source, multi-mode comparison, this scheme was extended with the two categories introduced above, giving seven mutually exclusive categories in a fixed precedence order: technical origin is checked before hallucination, so that a value matching real technical metadata is never double counted as a hallucination, followed by hallucinated, inferred (inference mode only), type mismatch, wrong value, incomplete, and correct. Three per-paper accuracy metrics were computed from these counts. `judge_accuracy` is the fraction of correct annotations over the total, the strict, text-only rate that is comparable between strict and inference mode as a sanity check, since it excludes inferred values in both. `judge_accuracy_adjusted` adds technical-origin annotations to the numerator. `judge_accuracy_with_inference` additionally credits inferred values in inference mode, giving the effective accuracy once expert-level inference is credited.

Reproducibility of the LLM judge. Because the judge is itself a generative model, its verdict on a given value can vary between calls. To quantify this, the complete judging pipeline, all three sources, both modes, and all 30 papers, was run three independent times against identical input data, each with its own response cache so that every call was answered fresh rather than replayed. Per-field and per-paper agreement was computed as the proportion of values for which all three runs returned the identical error category or verdict, and run-to-run noise was summarized as the standard deviation of `judge_accuracy` across the three runs, overall and broken down by source, judge mode, and metadata field. Fields with fewer than 10 judged rows for a given source were excluded from per-field reproducibility estimates as too sparse to give a stable rate.

Technical pipeline integration

LLM extraction operates on manuscript text and is therefore limited to information that authors chose to report. To complement and verify manuscript-derived metadata, we integrated outputs from the METI technical pipeline (see Technical extraction pipeline section).

Priority hierarchy

Field resolution followed a three-tier hierarchy reflecting distance from the data itself. METI-derived values were assigned the highest confidence score (1.0), as they originate from the raw data. PRIDE dataset properties, which are controlled vocabulary terms attached to the PRIDE record, are assigned a score of 0.9. LLM-extracted values from manuscript text receive a score of 0.5 by default. When multiple sources were available, METI outputs were prioritized, PRIDE properties retained for provenance and LLM values used in absence. These scores serve both as the confidence value reported for single-source fields and as the input to conflict resolution when multiple sources are available (see Conflict resolution). This hierarchy is

encoded in a field-to-getter mapping (PRIDE_TOOL_MAP) linking each field to its corresponding METI and PRIDE accessors.

Conflict resolution

Each field is resolved by comparing up to three sources: the LLM-extracted value, the METI-derived value, and the PRIDE descriptor. Two values are considered in agreement if they match exactly, one is a substring of the other, or their fuzzy similarity is at least 0.7 using the Ratcliff-Obershelp algorithm (Python's `diffib.SequenceMatcher`), which additionally handles binomial abbreviation patterns such as "P. falciparum" matching "Plasmodium falciparum".

When the LLM and METI values agree, the field is labeled AGREE and the METI value is retained, with confidence equal to the METI score. When they disagree and no PRIDE value is available, the field is labeled DISAGREE and the METI value is retained by default. When they disagree and a PRIDE descriptor is also available, resolution follows a three-way majority vote rather than a fixed hierarchy: if the LLM and PRIDE values agree with each other while METI is the outlier, the field is labeled DISAGREE_TOOL_OUTVOTED and the LLM value is accepted, with confidence 0.85; if METI and PRIDE agree with each other while the LLM value is the outlier, the field is labeled DISAGREE_LLM_OUTVOTED and the METI value is retained; if all three values differ from one another, the field is labeled DISAGREE_SPLIT, the METI value is retained, and its confidence is discounted by a factor of 0.8. (If the fuzzy threshold lets the LLM and METI values each independently agree with PRIDE despite disagreeing with each other directly, the field is likewise labeled AGREE.) Fields with a value from only one source are labeled LLM_ONLY, with confidence fixed at 0.5, or METI_ONLY, with confidence equal to the METI score; fields with no value from any source are labeled UNKNOWN.

Consequently, METI output does not automatically take precedence on disagreement: when the independently sourced PRIDE descriptor sides with the LLM against METI, the LLM value is accepted instead. Disagreements are logged with all available values, source attribution, and resolution rationale in a batch-level log, separate from individual outputs.

Pipeline only metadata

The integration step also appended metadata unavailable from manuscript text, including PTM-Shepherd modification statistics, SAGE quantification methods, acquisition metrics from `runAssessor`, PRIDE annotations, and provenance information (PXD, PMID, pipeline version). These fields are written into a `_meti_data` key in the enriched output and surfaced where relevant.

Benchmarking

This section evaluates the output of the full pipeline by comparing extracted and normalized metadata against SDRF-derived ground truth

Hierarchical semantic matching

Exact string matching between free-text LLM output and SDRF-derived ground truth systematically underestimates accuracy, as equivalent concepts are frequently expressed with different surface forms. For example, a mass spectrometer recorded as "Q Exactive HF-X" in SDRF may appear as "Q Exactive HF" in a manuscript; "trypsin" and "Trypsin/P" refer to the same cleavage agent but differ in SDRF encoding conventions; "liver" and "hepatic tissue" are semantically equivalent but would fail an exact comparison. To address this, we implemented

a five-level hierarchical matching strategy, applying increasingly permissive comparison criteria until the first match is found.

Level 1: Exact match (rank 1) compares strings character-for-character. This captures cases where SDRF terms are reproduced verbatim, which is common for controlled vocabulary fields such as sex, acquisition method, and numeric replicate counts.

Level 2: Normalized match (rank 2) applies case-folding and whitespace trimming. This handles the most common surface variation: capitalization differences between SDRF (e.g., "Label-free quantification") and LLM output (which may produce "label-free quantification").

Level 3: Ontology match (rank 3) resolves both values to ontology identifiers using the SapBERT-normalizer. Identical ontology IDs are considered equivalent regardless of surface form, capturing synonyms such as "Homo sapiens" and "human", or "Orbitrap Fusion Lumos" and "Lumos".

Level 4: Hierarchical ontology match (rank 4) allows credit when values map to different ontology nodes but share an ancestor in the DAG. This matching is bidirectional, either value might be broad, and score decays with graph distance never exceeding 4 hops in the graph. This is applied to fields with a defined ontology namespace in FIELD_ONTOLOGY_MAP (Cell Ontology, CL, UBERON, Mondo). For cell type fields an additional fallback is applied: if the LLM value matches a known cell line name in a curated lookup table, it is first mapped to its canonical cell type (e.g., "HeLa" → "epithelial cell", "Jurkat" → "T cell") before being passed to SapBERT, handling the common case where an LLM identifies a cell line that the SDRF records at the cell type level.

Level 5: Semantic match (rank 5) is applied when no prior level matches. Cosine similarity between SciBERT embeddings is computed using SentenceTransformers with L2 normalization, making dot products equivalent to cosine similarity. SciBERT is used due to its biomedical pretraining, improving representation of proteomics terminology. Embeddings are cached within each benchmark run to avoid redundant computation. A similarity at or above 0.75 is recorded as a SEMANTIC match, with the similarity value used directly as the score. Pairs below 0.75 are recorded as NO_MATCH, with the similarity value retained for diagnostic purposes.

Acknowledgments

Data Availability

Authorship Contribution Statement

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